# Who Benefits from Benefits?
# Municipal Consequences of Federal Cash Transfers

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**Abstract**

Centralized welfare programs limit local discretion over eligibility and payments, yet access often depends on municipal bureaucracies. We study whether local politicians obtain electoral returns from a federally designed cash transfer when their remaining discretion lies in registration. In Brazil's *Bolsa Família*, a 2009 reform unexpectedly changed the formula assigning municipal beneficiary quotas, and a later federal revision opened additional slots during the same mayoral term. We measure the cumulative authorization change relative to the allocation implied by the previous rule. In the 2,078 municipalities where the mayor elected in 2008 sought reelection in 2012, a one-standard-deviation quota shock created 171.50 additional authorized slots per 10,000 residents and induced 54.68 additional beneficiary families per 10,000 residents. The 2SLS estimate implies that 100 additional beneficiary families per 10,000 residents raise vote share by 2.59 percentage points. Enrollment expanded both with and without a pre-existing local welfare center (*CRAS*), but we find a clear electoral response only in *CRAS* municipalities. Within that group, the vote-share response is larger where offices report active outreach and more intensive service to *Bolsa Família* families. We interpret these patterns as scope conditions for the local implementation channel. Centralizing eligibility and payment does not eliminate local political returns when program access still depends on municipal implementation.

**Keywords:** cash transfers; electoral returns; state capacity; *Bolsa Família*.
**JEL Codes:** D72, H53, H75, I38, O17.

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# 1 Introduction

Across the developing world, cash transfer programs have become a leading tool for reducing poverty while insulating redistribution from local political discretion. By standardizing eligibility rules, fixing benefit values by federal decree, and processing payments through centralized banking systems, these programs are designed to reduce the scope for patronage and discretionary allocation (Fiszbein and Schady, 2009). Yet access often depends on local bureaucracies controlled by elected officials. Can local politicians benefit electorally from policies designed and financed by another level of government? This is the question we address.

We study this question in the context of Brazil's *Bolsa Família*, one of the largest conditional cash transfer programs in the world.[1] Although eligibility rules, benefit size, and payments are determined by the federal government, municipalities remain responsible for identifying and registering eligible households, mostly through local welfare centers known as *Centros de Referência de Assistência Social* (*CRAS*). This institutional arrangement creates a narrow but important margin of local discretion: while mayors cannot alter transfer amounts, redirect funds, or withhold payments, they control the administrative pipeline through which households may gain access to the program.

This margin depended on a registration process that required active municipal effort. Identifying potentially eligible families meant locating them, conducting household interviews, and collecting documentation; municipal authorities controlled the local registration apparatus, which gave them influence over the pace, intensity, and reach of enrollment efforts.[2] In practice, two municipalities facing similar levels of underlying need could generate very different realized coverage depending on how actively the local administration pursued registration.

Our empirical strategy starts from a 2009 reform that revised the formula used to compute municipal *Bolsa Família* quotas. The program operates through federally determined municipal quotas, which set a ceiling on the number of families that can be served in each municipality.[3] A mayor elected in October 2008 could expect the

[1] By June 2006, *Bolsa Família* covered 11.10 million families, approximately 46 million people, or about 25% of the Brazilian population. By June 2015, the program supported 13.80 million families, with an average monthly benefit of R$167.15 and a budget equivalent to 0.50% of GDP.

[2] Hellmann (2015) describes enrollment as a municipal responsibility. He reports that as of September 2014, 77% of registrations were made without a home visit. Municipalities were responsible for preparing the local structure for registration, training personnel, promoting awareness campaigns, and designing *active search* strategies to reach eligible families outside the existing registry.

[3] Lindert et al. (2007) explains that *Bolsa Família* maintained municipal program quotas based on local

next adjustment to follow the previous allocation rule. Instead, the federal government changed that rule in 2009 and revised municipal quotas again in 2012.[4] These revisions expanded the ceiling, not the registry. Families who were already registered could enter when new slots opened, while filling the remaining slots depended on local governments' effort to identify, locate, and register eligible families not yet in the federal database.

We combine administrative data on quotas, realized beneficiary counts, and local social-assistance infrastructure with municipal electoral data from the subsequent election. The construction of the 2009 benchmark follows Gerard et al. (2025): we apply the previous allocation rule to the information available before the reform. Our main measure subtracts that benchmark from the official 2012 quota, capturing the cumulative authorization that opened during the mayor's term. We use this quota change to instrument realized beneficiary expansion.

Our findings trace the reform along four stages: the relationship between federal authorization and realized enrollment; the electoral response to the authorization shock; the 2SLS estimate that combines the two; and the institutional conditions under which these patterns emerge.

The first stage quantifies how federal authorization translated into realized enrollment. A one-standard-deviation increase in the quota shock corresponds to 171.50 additional authorized beneficiary-family slots per 10,000 residents. It raises realized beneficiary expansion by 0.30 standard deviations, equivalent to 54.68 additional beneficiary families per 10,000 residents.

Once federal authorization is shown to translate into realized enrollment, we estimate the electoral return to expanding the ceiling itself, before separating out how much of that ceiling municipalities chose to fill. For the same one-standard-deviation quota shock, the reduced-form intention-to-treat estimate is 1.41 percentage points on the incumbent's vote share. This is the electoral consequence of additional federal authorization, integrating over the local enrollment response documented in the first stage.

Scaling the intention-to-treat effect by the first stage gives the electoral response

poverty estimates. Original municipal allocations were constructed by combining the 2001 PNAD with the 2000 Census, generating an initial national target of 11.20 million families when the program was created in 2003.

[4] Hellmann (2015) notes that *Bolsa Família* administered municipal quotas using estimates of poor families developed in partnership with IBGE and reports that, in 2009, the federal government refined the measurement of the target population by incorporating methodologies to capture income volatility among vulnerable households. These studies formed a major expansion of coverage, incorporating more than 1.60 million additional families.

to realized beneficiary expansion. A one-standard-deviation increase in instrumented enrollment, about 181.36 additional beneficiary families per 10,000 residents, raises incumbent vote share by 4.69 percentage points. Equivalently, an increase of 100 beneficiary families per 10,000 residents raises vote share by 2.59 percentage points. The estimate is similar across alternative quota definitions, outcome definitions, and corrections for selection into the rerunning sample.

These returns differ across the institutional layer that converts authorization into delivery. Of the 2,078 municipalities in the estimation sample, 1,394 had a pre-existing local welfare center in 2008 and 684 did not. Realized enrollment expands in both groups: the first-stage coefficients are 0.30 with a *CRAS* and 0.32 without one. The divergence is electoral. The corresponding 2SLS estimate is 6.11 in the *CRAS* group; outside it, the confidence set includes zero. These estimates locate a clearer electoral response in municipalities with established offices, although balancing observed municipal characteristics substantially reduces that contrast.

Within these municipalities, we next examine two measures of how the local welfare apparatus operated. The first records whether municipal teams proactively searched for potentially eligible families (active outreach) rather than waiting for them to appear at a registration point. The first stage remains positive both with and without this practice, while the reduced-form vote-share point estimate is 2.00 percentage points with active outreach and 0.76 without it. The second measure records whether the *CRAS* reported serving *Bolsa Família* families through PAIF (*Serviço de Proteção e Atendimento Integral à Família*), a continuing family-support service provided through the social-assistance system. The corresponding reduced-form estimates are 2.25 percentage points where at least one *Bolsa Família* family was served through PAIF and 0.65 where none was reported. The PAIF contrast becomes sharper when we consider service volume: among municipalities reporting at least one family served through PAIF, the reduced form is 5.36 above the median number of families served and indistinguishable from zero below it. Because both operational measures predate the quota shock but were not randomly assigned, we interpret these results as scope patterns. They motivate a contact interpretation, but subsequent tests of operating duration and reported list-use frequency do not establish a consistent electoral gradient. Comparisons across office-opening cohorts are descriptive, and the evidence leaves the specific CRAS mechanism unresolved.

These findings advance three related literatures. We speak first to the political economy of cash transfer programs. Studies of social transfers find gains in government support and, in some settings, incumbent voting, although electoral results vary across de-

signs and contexts (Manacorda et al., 2011; De La O, 2013; Imai et al., 2020; Frey, 2019). Much of this literature studies settings in which local discretion is deliberately minimized, relying on centrally determined thresholds, randomized assignment, or other exogenous variation to identify electoral effects. We complement this work by studying a setting in which the program architecture is centralized but the last mile of delivery is not. Our findings suggest that centralizing transfer rules and payment flows is not sufficient to depoliticize redistribution when access to the program depends on a locally controlled registration bureaucracy. Even when mayors cannot alter eligibility criteria or benefit levels, control over the administrative pipeline through which households gain program access can still generate measurable electoral returns.

Our paper is especially closely related to Frey (2019), but the two designs study electoral returns under different implementation regimes: Frey's intervention leaves less room for mayoral involvement, whereas ours depends directly on municipal administration. Frey exploits a discontinuity in federal funding for the *Family Health Program*, a primary-care program in which teams follow a federally specified protocol of routine home visits to poor households. His information channel is embedded in those visits: the teams disseminate *Bolsa Família* eligibility information as part of their routine work, making enrollment less dependent on a separate effort by the mayor's welfare administration. We study the complementary margin. In our setting, converting expanded federal authorization into realized enrollment requires municipal welfare staff to identify, locate, and register eligible families. The designs also draw variation from different parts of the municipal distribution. Frey's threshold design is concentrated among small, low-development municipalities near a funding threshold, whereas the quota reform created more room for expansion in larger municipalities that had not yet absorbed earlier increases. Within his sample, coverage growth through a federally structured outreach channel reduces incumbency advantage and shifts the composition of challengers toward less clientelistic parties. Within ours, coverage growth through a locally managed registration channel produces positive returns for the incumbent.

Second, we contribute to the literature on political accountability under shared governance. A central question in this literature is whether voters can attribute responsibility when multiple levels of government jointly produce policy outcomes (Przeworski et al., 1999; Besley, 2006). Ferraz and Finan (2008) show that voters punish corrupt mayors when randomly assigned audits reveal misuse of federal funds, especially where local media facilitate information transmission. Avis et al. (2018) show that audits also reduce corruption through a legal disciplining channel. Our setting differs because the

relevant outcome is positive delivery rather than malfeasance. The results are consistent with voters giving local incumbents some credit for the expansion of a federally funded program when access still depends on municipal action. We do not observe voters' beliefs directly, but the pattern fits an attribution channel in a setting of shared responsibility.

Third, we contribute to the literature on state capacity and the bureaucratic foundations of policy delivery. Besley and Persson (2009, 2011) establish state capacity as a precondition for credible policy delivery, and Finan et al. (2017) survey the personnel and bureaucratic infrastructure through which public programs are implemented. Our findings distinguish enrollment pass-through from political conversion. Municipalities both with and without a pre-existing *CRAS* translate the quota shock into additional beneficiaries. The initial electoral contrast favors municipalities where the delivery apparatus was already present, but is sensitive to their observed composition. This distinguishes the evidence for local electoral returns from the more demanding claim that welfare offices themselves produce those returns.

The remainder of the paper is organized as follows. Section 2 describes the institutional setting of *Bolsa Família*, the quota system, the federal revisions, and the data sources. Section 3 presents the impacts of the reform and summarizes the identification strategy, while the full empirical strategy appears in Online Appendix A and the figures and tables are collected after the references. Section 4 examines local implementation as a possible mechanism, evaluates the institutional comparisons, and discusses alternative explanations. Section 5 concludes.

# 2 Institutional Background and Data

## 2.1 *Bolsa Família* Program

**Program Design.** Launched in 2004, *Bolsa Família* consolidated several pre-existing social-assistance programs into a single conditional cash transfer system and became Brazil's flagship poverty-alleviation policy. In 2009, the program classified eligible families into two income groups. Families with monthly per capita income at or below R$70 were classified as extremely poor and received a basic monthly benefit of R$68, plus variable benefits of R$22 per child or adolescent aged 0–15, up to three, and R$33 per adolescent aged 16–17, up to two. Families with monthly per capita income above R$70 and at or below R$140 qualified only for the child- and adolescent-linked variable benefits. These benefits were conditional on age-specific education and health require-

ments, and the maximum monthly transfer per family was R$200. Table 1 summarizes the program's eligibility thresholds and benefit values from 2004 to 2018. By December 2009, the program reached approximately 12.40 million families, with an average benefit of about R$95.

**Federal Rules and Local Responsibility.** *Bolsa Família* combined centralized rules and payments with decentralized administration. The federal government, through the *Ministério do Desenvolvimento Social* (MDS), set eligibility criteria and benefit values, used *Cadastro Único* records to identify families eligible for payment, and transferred benefits directly through the state-owned bank *Caixa Econômica Federal*. State governments coordinated implementation across social assistance, education, and health and provided technical support to municipalities, but neither selected beneficiaries nor disbursed transfers.[5]

Municipal governments operated the program locally. They identified low-income families, collected and updated their registry information, conducted household visits, monitored compliance with education and health conditions, and coordinated the municipal agencies involved in implementation. Mayors therefore could not change eligibility thresholds or benefit values, redirect federal transfers, or guarantee that a registered family would receive a benefit. Their administrations could, however, affect how intensively potential beneficiaries were sought, how quickly applications were processed, and how completely eligible families became visible to the federal system. This separation between federal benefit rules and local implementation is the institutional margin central to our analysis.

**Municipal Registration.** The *Cadastro Único* (*CadÚnico*) is a national registry of low-income families and the administrative gateway to *Bolsa Família* and many other social programs. It covered families with per capita income below half the minimum wage (R$232.50 in 2009), well above the program's own eligibility thresholds. Registration was therefore necessary but not sufficient for benefit receipt, and the registered population was substantially larger than the beneficiary population. Under Decree No. 6,135/2007, the MDS defined and supervised the national registry while participating municipalities collected family information using a standardized form.

Registration required a face-to-face interview covering household composition,

[5]The division of responsibilities during the study period was governed principally by Law No. 10,836/2004, Decree No. 5,209/2004, and MDS Ordinance No. 246/2005. The ordinance formalized municipal adherence to the program and specified federal and municipal commitments.

housing conditions, assets, employment, and income. In urban areas, families could seek a municipal registration post, often located at a *Centro de Referência de Assistência Social* (*CRAS*). In rural and remote areas, reaching a fixed office could be costly or impractical. Coverage there depended more heavily on active outreach: municipal teams had to locate potential beneficiaries, travel to underserved communities, and conduct interviews in or near their homes. Federal rules remained uniform, but the reach and timing of this local effort shaped which families entered the registry and when.[6]

## 2.2 Institutional Details of the Quota Reform

**Quotas and Enrollment.** Eligibility alone does not guarantee benefit receipt. Access to *Bolsa Família* depends on the availability of municipal quotas: caps on the number of beneficiaries per municipality, defined nationally and revised approximately every three years based on poverty estimates derived from census and survey data. These quotas function as official upper bounds on municipal enrollment, but the constraint is not always immediately binding. In practice, municipalities frequently operate near their ceiling, oscillating modestly above or below it from month to month before the federal priority mechanism reallocates unused slots.

**Priority and Local Effort.** The priority index considers both quota utilization and unmet registered demand: municipalities with low beneficiary-to-quota ratios and few eligible-but-unserved families risk seeing unused slots reallocated to places with stronger registered demand. The quota system therefore creates pressure to use the available slots. A quota increase is not a benefit guarantee, but it opens federally authorized room that municipalities must fill by identifying, registering, and updating eligible households. Effective coverage thus depends on local administrative initiative: political will and organizational capacity determine how much of the authorized ceiling becomes actual benefit receipt.

**Quota Reform.** In 2009, the federal government revised the methodology used to compute municipal quotas, producing a heterogeneous reallocation of authorized ben-

[6] A *CRAS* is the local public facility responsible for basic social-assistance services and a visible point of contact between families and municipal government; it is not the only place where *CadÚnico* interviews may occur. The 2009 *Censo SUAS* recorded 5,798 units in 4,329 municipalities. Municipalities without a *CRAS* could still register families through other municipal facilities or outreach teams, so absence of a unit does not imply absence of registration capacity.

eficiary slots across municipalities. The previous approach, in place since 2006, distributed slots in proportion to each municipality's estimated number of poor families from the 2000 Census, adjusted for intercensal population growth. The 2009 revised methodology accounts for income volatility among families near the eligibility line, a feature of transient poverty that earlier projections systematically undercount (Gerard et al., 2025).

**Reallocation Shock.** Alongside the methodological update, all municipal quotas were scaled by a factor of 1.18 to account for structural income volatility near the poverty threshold. Because this multiplicative factor is constant, it contributes no cross-municipal variation to the instrument; all identifying variation in the quota shock derives from the updated municipal poverty estimates and the change in income thresholds. Together, these changes added nearly 1.60 million new authorized slots in 2009. The reform expanded the ceiling, not the registry: although already-registered eligible families were absorbed automatically, the magnitude of the expansion left a large residual margin of unfilled slots whose realization depended on the municipal government's effort to identify, locate, and register eligible families not yet in the federal database.

## 2.3 Data

Our empirical analysis combines five sources of data: administrative records on *Bolsa Família*, family-level records from the *Cadastro Único*, electoral returns, census and household-survey data, and administrative records on local social-assistance infrastructure.

***Bolsa Família* Administrative Records.** We begin with monthly administrative records from the *Ministério do Desenvolvimento Social* on *Bolsa Família* quotas, realized coverage, and total transfers from 2006 to 2012. These data span the pre-reform baseline, the 2009 reallocation, and the post-reform expansion window through the 2012 election, and allow us to measure both authorized and realized program expansion.

***Cadastro Único*.** We also use family-level records from the 2012 *Cadastro Único*, provided by the *Ministério do Desenvolvimento Social*. The file covers families recorded in the registry in 2012, including those registered in earlier years. It reports their municipality of residence, per-capita family income, family characteristics, and dates of first

registration and last update. The first-registration date identifies entry into the registry separately from subsequent updates. We use it to construct registration cohorts, including families first registered before the reform.

**Snapshot and Updating.** Income and family characteristics reflect the information on file in 2012. During the study period, registry information required updating or revalidation every two years.[7] A family first registered in 2007 can therefore be assigned to its registration cohort using the 2012 file. Its recorded income, however, describes its circumstances as last reported by 2012. The cohort comparisons describe families observed in that snapshot, grouped by first-registration date and income recorded in 2012. Families absent from the snapshot and their earlier income histories remain outside this exercise.

**Electoral Records.** We then use electoral records from the *Tribunal Superior Eleitoral* (*TSE*) for the 2008 and 2012 municipal elections. These data provide candidate-level vote totals, turnout, and detailed candidate characteristics, including education, prior occupation, and party affiliation.

**Census and Household Surveys.** To reconstruct counterfactual municipal quotas under the pre-reform allocation rule, we combine the 2000 Census from the *Instituto Brasileiro de Geografia e Estatística* (*IBGE*) with poverty estimates from the 2006 *Pesquisa Nacional por Amostra de Domicílios* (*PNAD*). These are the inputs used to approximate the quota formula that would have prevailed without the 2009 reform.

**Censo SUAS.** Finally, we use administrative records from the *Censo SUAS* covering *CRAS* units in 2008 and 2009. These data describe local infrastructure, active outreach, and recorded numbers of beneficiary families served. We use the family count as a provisional proxy for PAIF activity, subject to the measurement limits discussed in Section 4.3. To examine office duration and network expansion, we also harmonize the annual censuses from 2007 through 2012. Absence from a census is treated as missing information rather than evidence that an office closed.

[7] Article 7 of Decree No. 6,135 of June 26, 2007 established the two-year validity of registry information.

## 2.4 Sample Construction

**Municipal Panel and Data Availability.** Our electoral analysis uses municipality-year observations for the 2008 and 2012 elections. We combine the sources using municipal identifiers and require observations in both years. The integrated database contains 5,568 municipal identifiers across election years. This is the starting universe for the sample accounting in Table 2. Requiring the inputs for quota shocks and realized enrollment excludes 63 municipalities, leaving 5,505. Missing records identify gaps in the measures needed for estimation. They do not establish that a municipality lacked access to *Bolsa Família*. Program exposure is measured in beneficiary families: the quota shock changes authorized family slots, and realized expansion measures the net change in beneficiary families at the municipal level.

**Electoral Sample and Record Quality.** We follow the mayor elected in 2008 and measure that mayor's vote share in each election. Requiring a candidacy in 2012 excludes 3,140 municipalities from the preceding universe and leaves 2,365. We then exclude 1 municipalities with incomplete electoral information and 83 municipalities with an uncontested election in either year. A further 84 municipalities lack a positive recorded vote share for the tracked mayor. Linking candidates across elections also requires an unambiguous name match, which excludes 7 municipalities. Removing 85 municipalities with multiple elected-status records leaves 2,105. We additionally check turnout identities and election validity. These checks impose no further losses in the displayed sequence. Table 2 reports every restriction, including those with no incremental exclusions.

**Final Estimation Sample and Selection.** Complete information on municipal transfers excludes 24 municipalities, and missing mayor characteristics exclude another 3. The final sample contains 2,078 municipalities and 4,156 municipality-year observations. We use this common sample across the main specifications. Its construction also requires the variables used in the alternative windows and mechanism comparisons. Table 3 compares it with a broader descriptive sample of 4,976 municipalities. The selection diagnostic uses a separate complete-case universe of 5,492 municipalities before conditioning on candidacy.

**Selection into Rerunning.** Selection matters because the reform could affect whether the incumbent runs again. In the preferred selection specification, a one-standard-

deviation quota shock predicts a 0.04 percentage-point change in that probability. Reweighting the electoral regression by the inverse probability of selection yields an estimate of 4.14 percentage points, compared with 4.69 in the main specification. These exercises provide evidence on selection through observed characteristics, while selection on unobserved determinants remains a limitation. Online Appendix A, especially Section A.3.1, presents the selection tests and their uncertainty.

## 2.5 Descriptive Statistics

**Municipality and Mayor Characteristics.** Table 3 describes municipalities and their elected mayors in the broader comparison sample and the estimation sample. The 2008 winner's vote share averages 54.97 percent in the former and 52.92 percent in the latter. Predetermined municipal characteristics, including urbanization, poverty, income, and inequality, have similar means across the two panels. Mayor characteristics differ more, including prior occupation and age. The estimation sample describes incumbents who contest another election, so these differences help delimit the population studied. Similar municipal averages alone do not establish that selection is unrelated to electoral outcomes.

**Program Coverage and Expansion.** Municipalities in the estimation sample had an average of 879.87 beneficiary families per 10,000 residents in 2008. Realized coverage rises by 203.50 families per 10,000 residents on average through 2012, with a median increase of 163.15. The realized measures use twelve-month averages of monthly beneficiary-family counts. Rates use municipal population in 2006 as a fixed denominator, so changes reflect enrollment rather than contemporaneous population growth. Authorized and realized expansion vary across municipalities and are right-skewed. We express regression coefficients per one analysis-weighted standard deviation to make their scale interpretable and translate the estimates into families per 10,000 residents. These measures count families receiving benefits, whereas the *Cadastro Único* cohort exercise describes registered families using the 2012 snapshot.

**Local Social-Assistance Infrastructure.** The estimation sample includes 1,394 municipalities with a pre-existing *CRAS* and 684 without one. Municipalities with a *CRAS* account for 67.08 percent of the sample. The infrastructure measures also distinguish whether these units conduct active outreach and provide family-assistance services. These characteristics describe the local capacity through which eligible families may en-

ter the registry and obtain benefits. Section 4 examines whether the electoral response to expansion differs across these settings. That comparison conditions on existing infrastructure and does not identify the effect of installing a *CRAS*.

Figures 1, 2, and 3 collected after the references complement these statistics with the evolution of quota utilization, the relationship between authorized and realized enrollment, and the distribution of unused authorization. The tables discussed below are collected there as well.

# 3 Impacts of the *Bolsa Família* Reform

Municipal incumbents gain an estimated 4.69 percentage points of vote share following a one-standard-deviation expansion in local enrollment induced by federal authorization (Anderson–Rubin 95 percent set: [1.24, 10.28]). We begin by documenting how federal authorization changed municipalities' room to enroll families. We then ask whether that authorization translated into electoral support, and whether the answer depends on realized enrollment. The federal government sets the program's quotas and pays its benefits, while municipal offices identify and register eligible families. Our estimates concern mayors who sought reelection and the enrollment expansion induced by federal authorization. The main tables and figures appear at the end of the text, before the appendix.

## 3.1 Policy-Induced Variation in Authorization and Coverage

**Authorization.** The reform created variation in how many families municipalities were authorized to enroll during the mayoral term. The 2009 reform changed that authorization, and further quota revisions followed in 2012. We measure cumulative authorization as the official 2012 quota minus the counterfactual quota that the previous allocation rule would have implied in 2009, evaluated at the R$90/137 poverty thresholds. The comparison holds the earlier rule as the benchmark and follows enrollment through the end of the term. One standard deviation of the authorization shock corresponds to 171.50 additional authorized families per 10,000 residents.

**Realized Enrollment.** Did authorized capacity translate into realized coverage? Municipalities enrolled more families where the federal government authorized greater expansion, across the reported control sets. An increase of one standard deviation in

authorization brings an estimated 54.68 more beneficiary families per 10,000 residents (Table 4). Each additional authorized place corresponds to 0.32 additional families receiving the benefit.[8] This is where municipal implementation enters: federal authorization opens places, while local staff identify and register eligible families who may fill them.

## 3.2 Identification Strategy

Mayors may expand enrollment in response to electoral pressure, and municipalities differ in their ability to reach eligible families. A direct comparison of enrollment growth and votes would combine these local choices with the political consequences of the program. We instead compare changes in incumbent support between municipalities exposed to different federal authorization shocks, conditional on pre-reform characteristics and incumbent party.

Let $\mathbf{D}$ collect municipalities' enrollment changes, $\mathbf{Z}$ their authorization shocks, and $\Delta\mathbf{Y}$ their changes in incumbent vote share. The matrix $\mathbf{X}^0$ contains pre-reform controls, including incumbent party. We estimate

$$\mathbf{D} = \pi_0\mathbf{1} + \pi_1\mathbf{Z} + \mathbf{X}^0\boldsymbol{\pi}_2 + \boldsymbol{\eta}, \tag{1}$$

$$\Delta\mathbf{Y} = \beta_0\mathbf{1} + \beta_1\widehat{\mathbf{D}} + \mathbf{X}^0\boldsymbol{\beta}_2 + \boldsymbol{\varepsilon}, \tag{2}$$

where $\widehat{\mathbf{D}}$ contains the first-stage fitted values and $\mathbf{1}$ is a vector of ones. We estimate turnout in the same way. Enrollment and authorization are rates per 10,000 residents using fixed 2006 population, divided by their respective analysis-weighted standard deviations on the common sample. Electoral responses are reported in percentage points. Throughout, intervals accompanying IV estimates are Anderson–Rubin 95 percent sets.

We weight municipalities by 2006 population, so the estimates give greater influence to places where more people live. Standard errors are clustered by state. With two periods and fixed municipality weights, equations (1) and (2) give the same point estimates as the municipality and year fixed-effects panel reported in the tables.

**Validity of the Design.** The authorization comparison requires the quota shock to be unrelated to the electoral trend that would otherwise have occurred, conditional on the

[8] On the standardized scale, the first-stage coefficients are 0.32, 0.36, and 0.30 as the control set expands. Enrollment growth compares the twelve-month average family counts in 2008 and 2012, allowing for entries and exits. These coefficients describe net expansion, not an individual family's probability of receiving a benefit.

controls. Interpreting the enrollment coefficient additionally requires authorization to affect votes through realized enrollment and a continuous-treatment monotonicity condition. Online Appendix A develops these assumptions and examines selection into rerunning, pre-reform electoral balance, alternative exclusion channels, and first-stage monotonicity. The diagnostics help assess the design; they cannot establish exclusion or eliminate selection through unobserved characteristics.

## 3.3 Electoral Responses

**Authorization and Incumbent Vote Share.** The authorization comparison asks whether mayors receive more votes where the federal government permits greater expansion. The preferred reduced-form estimate is 1.41 percentage points per standard deviation of the quota shock (Table 5). With the preceding control sets, the estimates are 1.93 and 1.30. This comparison captures the electoral response to authorization, including any response that operates before families actually receive benefits.

**Realized Enrollment and Incumbent Vote Share.** The electoral return to coverage is 4.69 percentage points of incumbent vote share for a one-standard-deviation expansion in local enrollment ([1.24, 10.28]; Table 6). The estimated electoral gain can be distinguished from zero, although its magnitude remains uncertain. To put the change in terms of families served, an additional 100 beneficiary families per 10,000 residents corresponds to an estimated 2.59 percentage points of vote share ([0.68, 5.67]).[9] For comparison, the median incumbent lost 3.03 percentage points of vote share between these elections. The estimate describes a change in the mayor's electoral support at the municipal level; it does not reveal which voters changed their vote or how they assigned credit.

**Turnout.** Turnout provides a separate margin of electoral response. We do not find a clear turnout response to local program expansion; the confidence set includes zero. The preferred response to authorization is -0.35 percentage points, and the response to instrumented enrollment is -1.17 ([-2.79, 0.05]). Tables 5 and 6 report both outcomes under the same control sequences. Separating turnout from incumbent vote share matters because receiving more votes and bringing more voters to the polls are distinct political responses.

[9] One standard deviation of enrollment growth is 181.36 beneficiary families per 10,000 residents. The natural-unit coefficient and interval rescale the same IV estimate.

## 3.4 Heterogeneity and Scope

**Municipality Size.** The pooled response is concentrated in municipalities of a particular size. The electoral gains are clearest among mayors of larger municipalities. The confidence sets exclude zero in the two upper population tails, while the evidence for the two lower-population samples remains inconclusive. Population weighting makes this distinction relevant to the interpretation of the pooled result. Table 7 reports estimates for municipalities below the median, below the 75th percentile, and in the two upper population tails.[10] These comparisons show where the estimated response appears in the sample. They do not identify the effect of making a municipality larger.

**Baseline Coverage and Prior Expansion.** Prior coverage may leave less room for the authorization shock to operate. The conditional estimates point to electoral gains below median baseline coverage and remain inconclusive above the median. Below and above median baseline coverage, the estimates are 10.19 ([1.72, 20.80]) and -0.41 ([-1.84, 0.81]), respectively (Table 8). These paired comparisons use conditional inference, which limits what we can conclude about differences between the groups.[11]

The same table compares municipalities inside and outside the Frey (2019) sample. The outside and inside estimates are 14.20 ([3.87, 52.46]) and -0.12 ([-1.71, 1.48]). The conditional estimates point to electoral gains outside the Frey sample and remain inconclusive inside that sample. The conditional first-stage $F$ statistics are 9.75 and 44.76, respectively. This comparison relies on the enrollment response within each group. Earlier expansion could have reduced the eligible pool still available for enrollment. But coverage in 2008, although measured before our shock, may reflect that earlier expansion as well as other municipal differences. The comparison cannot establish that the program had exhausted the eligible pool or that saturation explains the electoral pattern.

**Region and Party.** The regional and partisan comparisons locate the pooled return across municipalities and incumbents. Outside and inside the Northeast, the estimates are 2.71 ([-1.79, 9.13]) and 1.01 ([-0.73, 2.60]; Table 9). The regional comparison does

[10] Below the median and below the 75th percentile, the IV estimates are 0.07 ([-1.44, 1.37]) and 0.11 ([-1.47, 1.79]). Above the 75th percentile, the reduced form is 2.91 and the IV estimate is 11.27 ([3.37, 27.32]). Above the 90th percentile, they are 5.42 and 18.26 ([8.68, 60.48]).

[11] The baseline-coverage, Frey, region, and party partitions allow instruments, endogenous variables, and the complete control specification to differ across groups. The joint regression retains their covariance. The reported AR sets hold the other group's coefficient at its estimate; they are conditional diagnostics rather than joint confidence regions robust to weak identification of both coefficients. Differences use conventional state-clustered tests and are not robust to weak instruments.

not establish a difference in the electoral return under the conventional state-clustered test at the five-percent level. For non-PT and PT incumbents, the estimates are 6.49 ([1.67, 13.10]) and -1.37 ([-10.59, 13.50]). The conditional estimates point to electoral gains for non-PT incumbents and remain inconclusive for PT incumbents. The party-group comparison does not establish a difference in the electoral return under the conventional state-clustered test at the five-percent level. These comparisons describe the reach of the pooled finding; they do not identify how voters divide credit between federal and local politicians.

## 3.5 Robustness

**Outcome Definitions and Extreme Values.** The electoral return could depend on how we measure support or on a few extreme observations. Table 10 addresses both concerns. Winsorizing vote share gives an estimate of 4.23 ([1.16, 9.21]), while using registered voters as the denominator gives 3.80 ([0.91, 8.41]). Winsorizing the authorization shock gives 5.50 ([1.78, 11.01]). The registered-voter measure counts support relative to the electorate, rather than only voters who cast a valid vote for a candidate.

**Authorization Horizon.** The 2012 quota revision may matter for the expansion observed over the mayoral term. Using only the official 2009 quota relative to the counterfactual allocation gives an estimate of 2.47 ([-3.25, 7.85]). Enrollment still measures the full 2008–2012 change. This comparison asks whether the electoral result depends on the later authorization endpoint; it does not isolate the return to families enrolled during 2008–2009.

**Selection into Rerunning.** The sample contains mayors who chose to seek reelection. Reweighting for that choice using observed characteristics gives an estimate of 4.14 ([0.98, 8.81]). This adjustment combines inverse-probability and population weights. Its value depends on overlap and on the selection model, which cannot account for unobserved determinants of both rerunning and electoral performance.

**Normalization and Discretization.** We also examine whether the result depends on the way municipal expansion is measured. Using Census-2000 families as the denominator for enrollment and authorization gives 4.60 ([0.77, 10.91]). This changes the definition of exposure while retaining population weights in estimation.

A separate exercise asks whether authorization moves municipalities across successively higher thresholds of enrollment growth (Table 11). The estimates for crossing the median, 75th-percentile, and 90th-percentile thresholds are 21.95 ([5.09, 67.37]), 17.65 ([5.44, 33.66]), and 22.45 ([6.67, 47.81]). The continuous specification gives 4.69 ([1.24, 10.28]). Because every panel keeps the same authorization instrument, the reduced form remains 1.41 by construction. The threshold estimates refer to different changes in program coverage than the continuous coefficient. Their causal interpretation requires exclusion through the chosen binary treatment, so they do not by themselves recover an unrestricted dose-response relationship.

# 4 Possible Mechanisms

Why might mayors gain votes when the federal government expands a transfer program? Helping families register may make the municipality's contribution visible, while accompanying services may give families additional reasons to support the incumbent. Income gains could also change political preferences or participation without families crediting the mayor for access to the program. Our data do not fully distinguish these explanations. We examine pre-existing welfare offices and their activities to assess whether contact with municipal staff helps account for the electoral gains, then consider the alternatives.

Families receive *Bolsa Família* payments from the federal government, but often encounter the program through municipal staff. Help with registration may lead families to associate a federal benefit with the local administration. The transfer literature also points to beliefs about government and reciprocity as possible sources of political support, without isolating the role of contact with program staff (Manacorda et al., 2011).

We use the quota-induced expansion from Section 3 to compare enrollment and electoral gains across municipalities with different welfare infrastructure.[12] Because offices and service practices were not randomly assigned, these comparisons can inform the contact explanation without identifying the effect of providing those services.

[12]The subgroup IV comparisons start from the main sample and retain its scaling, population weights and baseline controls, with further sample restrictions stated below, and allow the instrument, treatment and controls to vary by group. Standard errors are clustered by state. Group-specific Anderson–Rubin sets are conditional diagnostics that hold the other group's coefficient at its estimate; they are not joint confidence regions robust to weak instruments. Conventional difference tests accompany these comparisons. Additional reduced-form diagnostics report state-clustered intervals, wild-cluster tests and the specified multiplicity adjustments.

## 4.1 Local Welfare Offices and Electoral Returns

Quota expansion also increases first registrations in *Cadastro Único*. Table 12 shows that the response is concentrated among families whose recorded income places them within the program's income eligibility range. This is consistent with municipalities bringing potential beneficiaries into the registry. The data are a snapshot taken after the reform: income is measured in that snapshot, and first registration does not establish subsequent benefit receipt or identify which families changed their votes.

A *CRAS* gives families a place to seek assistance, meet municipal staff, and obtain help accessing social programs. Of the 2,078 municipalities in our sample, 1,394 already had a recorded center in 2008 and 684 had none recorded. We ask whether mayors gain more votes from program expansion where families already have this point of contact with the administration.

Federal authorization expanded enrollment in municipalities both with and without a welfare center. The first-stage coefficients are 0.30 with a *CRAS* and 0.32 without one. The data do not clearly distinguish the two enrollment responses. Table 13 reports both enrollment and electoral outcomes.[13]

The electoral gains are concentrated in municipalities that already had a welfare center. A one-standard-deviation increase in enrollment induced by the quota shock corresponds to 6.11 percentage points of incumbent vote share with a *CRAS*, compared with -1.23 without one. The authorization estimates follow the same comparison: 1.84 and -0.40 percentage points, respectively.[14]

This contrast directs attention to the municipality's role in families' access to the program. A welfare office may make that role more visible even when families in other municipalities also receive the federal benefit. The comparison is consistent with local contact helping voters associate access to the program with their mayor. Other explanations remain open: municipalities with a center may differ in the families reached by expansion, the services those families receive, or other features of local administration.

Municipal differences matter for this interpretation. Table 14 reweights municipalities with and without a baseline office to balance observed characteristics. The difference in electoral returns becomes substantially smaller under those weights, while enrollment still increases in both groups (Table 15). Reweighting changes which mu-

[13]The test of equal first-stage coefficients gives $p = 0.682$. A failure to distinguish these coefficients does not establish that the enrollment responses are identical.

[14]The conditional AR sets for the vote-share estimates are [2.12, 13.45] with a *CRAS* and [-5.25, 2.99] without one. The conventional test of their difference gives $p = 0.007$.

nicipalities receive the most emphasis and leaves unobserved differences uncontrolled. The sensitivity cautions against attributing the original electoral difference to the office itself.

## 4.2 Outreach and Family Contact

Outreach brings municipal staff to families who may not seek assistance at an office. It may make municipal effort visible to people who would otherwise have little contact with the welfare administration. Within municipalities that already had a *CRAS*, we compare those reporting active outreach before the reform with those without recorded outreach. The underlying coding groups missing responses with no recorded outreach, but an audit of the original source finds no missing outreach responses in this estimation sample.

The estimated electoral return is larger where offices report active outreach: 7.27 percentage points, compared with 1.74 without recorded active outreach. The estimates leave considerable uncertainty about the gap between these electoral returns. Table 16 reports the comparison.[15] The first-stage point estimate is smaller in municipalities reporting active outreach.

These offices were already conducting outreach before the reform. The comparison therefore cannot measure an outreach campaign caused by expansion. Their resources, the families they serve, or the visibility of local officials could also account for the difference in electoral returns.

**Reported Contact Practices.** If outreach helps families associate program access with the mayor, more frequent outreach might accompany larger electoral gains. The baseline census asks how often staff use lists to search for vulnerable families. We average the ordered responses across offices within each municipality and ask whether registration, enrollment and votes respond more strongly to quota expansion where this score is higher. Table 17 reports the results within municipalities with a baseline CRAS. For a standard-deviation increase in the score, the electoral response to a standard-deviation quota shock increases by an estimated 0.394 percentage points [-0.691, 1.479]. The corresponding change in enrollment is 4.035 additional beneficiary families per 10,000 residents [-5.587, 13.656]. Both estimates are imprecise, and neither survives the stated multiplicity adjustment.

[15] The conditional AR sets are [2.40, 13.66] with outreach and [-1.70, 7.41] without it. The conventional difference test gives $p = 0.051$.

Table 18 examines referrals to the registry, priority given to family follow-up, and an exploratory measure of outreach to families failing to meet program conditions. Table 19 uses an alternative coding of outreach frequency and allows quota responses to vary with other baseline municipal characteristics. These comparisons do not show a consistent link between reported contact practices and electoral gains. Referrals are almost universally reported, and population weights give relatively few municipalities much of the influence on the estimates. These staff reports measure neither completed registrations nor actual encounters or the credit voters assign to officials. The imprecision leaves room for meaningful contact effects.

## 4.3 Family Follow-Up through PAIF

Family follow-up through PAIF may keep beneficiaries in contact with municipal staff after they first seek assistance. Such services could give families repeated opportunities to observe the municipality's role. We use the pre-reform record of *Bolsa Família* families served by welfare offices as a proxy for PAIF activity, although this count does not measure how often each family meets staff.

The estimated electoral return is larger where offices record families served through PAIF: 7.70 percentage points, compared with 1.76 without a positive recorded family count. The estimates leave considerable uncertainty about the gap between these electoral returns. Table 20 reports this comparison.[16] The first-stage point estimate is smaller in municipalities reporting PAIF follow-up.

We also distinguish municipalities by the recorded number of beneficiary families served, summed across their welfare offices. This comparison uses the 962 municipalities with a positive recorded count, divided at their pre-reform median; it therefore concerns a narrower population than the preceding comparison. The estimated electoral return is larger in the higher-volume group: 18.78 percentage points, compared with -1.51 in the lower-volume group. The evidence supports a difference in electoral returns between these groups. Table 21 presents the estimates.[17] The recorded family count does not measure the quality of assistance or the amount of contact received by

[16]The conditional AR sets are [2.01, 15.78] with recorded PAIF follow-up and [-1.50, 9.53] without it. The conventional difference test gives $p = 0.095$. The conditional first-stage $F$ statistics are 38.04 and 8.18, respectively.

[17]The conditional AR sets are [8.01, 35.05] in the higher-volume group and [-7.71, 5.34] in the lower-volume group. The conventional difference test gives $p = 0.017$. The lower group includes observations at the median. Both PAIF comparisons use a family-count item that lacks its complete original questionnaire label in our extract. Its interpretation as PAIF follow-up remains provisional. The binary comparison codes missing counts together with zeros. The count is a proxy for service activity, not a verified measure of repeated contact.

each family.

## 4.4 Institutional Exposure and Network Expansion

We harmonize the official CRAS censuses from 2007 through 2012, retaining both the office counts observed in each wave and the reported opening dates.[18]

An office that has operated for longer may give families more opportunities to encounter municipal staff. A recent opening could instead make local action especially visible. We measure years of operation using opening dates reported before the reform. Table 22 asks whether quota expansion brings larger electoral gains where offices have operated for longer, allowing separately for whether a baseline office exists. The electoral response to a standard-deviation quota shock increases by an estimated 0.788 percentage points per additional year of operation [-0.049, 1.626], with adjusted $p = 0.952$. The estimate remains imprecise when we account for baseline office density or exclude dates that may refer to predecessor facilities. Table 23 compares offices established by 2006, those established in 2007–2008, and municipalities without a recorded baseline office.

The comparison between older and recent offices also depends on which municipalities receive the most weight. Within the baseline-CRAS sample, the older-minus-recent difference in the response to a standard-deviation quota shock is 1.444 percentage points [-0.154, 3.043] under population weights. After balancing the two groups on observed baseline characteristics, it is -0.436 [-2.036, 1.164] (Table 24). Unobserved differences remain, and reweighting changes the population emphasized by the comparison. The apparent advantage of older offices does not persist under these weights.

Table 25 and Figure 4 extend the comparison to offices reported as opening during the mayoral term. The late-entry group has a smaller electoral point estimate than the oldest baseline group: the difference per standard-deviation quota shock is -2.941 percentage points [-4.325, -1.558], with adjusted $p = 0.228$. The adjusted test does not establish this difference, and comparisons between successive opening cohorts do not show that electoral gains rise systematically with years of exposure. These groups reflect post-reform decisions about office placement, which offices survive, and which have valid reported dates. Their comparison remains descriptive. Later adopters remain in the main estimation sample, and their entry does not redefine its control group.

[18]The Ministry of Social Development provides the annual data, questionnaires and manuals through its Censo SUAS archive. Baseline duration uses dates reported in 2008, rather than dates recalled after the reform. Source questionnaires and manuals distinguish registry referral from registration and identify the list-use items used in the contact comparisons.

**Network Expansion and Identification.** An office can first appear in the census without having just opened, and a new identifier need not mean a new office. We treat an unobserved municipality as missing and exclude reported openings when an earlier census already records that office in operation. Table 26 checks whether municipalities with baseline offices continue to appear in later censuses. The adjusted tests do not establish that quota changes predict nonresponse. Missing census records still cannot verify closures or resolve uncertain opening dates.

Could quota expansion itself lead municipalities to open offices? This matters because new services might win votes independently of benefit receipt. Table 27 tests whether quota changes predict the observed expansion of the network. For a standard-deviation increase in the initial 2009 shock, the estimated change in the probability of observing an office in 2012 among municipalities coded without a baseline CRAS is 0.775 percentage points, with a confidence interval of [-0.564, 2.114]. The estimated response of date-consistent reported openings is -0.148 offices [-0.326, 0.031]. These estimates do not establish that quota expansion led to more offices. One association in the broader set of tests survives adjustment: a standard-deviation increase in the cumulative quota shock is associated with a change of -0.257 offices by 2010 [-0.413, -0.101], with adjusted $p = 0.046$. This quota measure includes changes after the 2010 census, so the association cannot establish their causal effect on earlier openings. It nevertheless prevents us from concluding that quota changes are unrelated to infrastructure. Services that respond to authorization and independently affect votes would challenge the interpretation of the IV estimate as an effect operating only through enrollment.

## 4.5 Alternative Explanations and Interpretation

**Material Benefits and Political Credit.** Additional income could change how families evaluate government, independently of the office that helped them register. In Uruguay, Manacorda et al. (2011) find persistent government support after transfers end and discuss both beliefs about government and reciprocity as explanations. Our municipal election data do not reveal these beliefs or whom families credit for assistance. We also cannot separate broader local economic effects on voters who do not receive transfers. Electoral gains could therefore arise without an encounter with municipal staff.

**Participation and Electoral Composition.** A mayor's vote share can rise because voters switch candidates, because turnout changes, or because new voters enter the elec-

torate. Table 28 expresses incumbent votes, other candidates' votes, blank or null ballots, and abstention using a common electoral denominator. It also reports registered voters relative to fixed baseline population. The registered electorate responds to quota exposure, while the CRAS differences in these outcomes do not survive their stated multiplicity adjustment. Municipal totals cannot reveal which individuals switched candidates, entered the electorate, or stopped voting. We therefore cannot establish persuasion rather than mobilization as the source of the mayor's gain.[19]

**Complementary Services and Political Relationships.** Offices that register families also provide assistance that families may value in its own right. The PAIF comparisons cannot separate appreciation of those services from credit for access to the federal transfer. Differences in who becomes enrolled could also explain why electoral gains vary across municipalities, even if contact has no independent effect. The sensitivity to municipal composition makes this possibility relevant. Political relationships may also change as families become less economically vulnerable: Frey (2019) interprets a negative local-incumbency response to transfer expansion through reduced opportunities for clientelistic exchange. Our positive estimate concerns a different source of expansion and population, and allows for offsetting changes in such relationships. We do not observe exchanges of benefits for votes.

Local contact and attribution remain plausible explanations for the electoral gains. The evidence does not establish that CRAS caused those gains, that longer operation strengthens them, or that any reported practice accounts for how expansion produces votes. The estimates for duration and contact are imprecise, the comparisons depend on municipal composition, and expansion of the office network remains a concern for identification. Distinguishing these explanations would require direct evidence on families' encounters and beliefs or independent variation in the services and information they receive.

# 5 Conclusion

Mayors can gain electoral support from a transfer program even when the federal government sets its rules and pays its benefits. In *Bolsa Família*, municipalities help families

[19]The distinction also arises in the Progresa literature: De La O (2013) advances a mobilization interpretation, whereas Imai et al. (2020) show that corrected geographic matches and electoral denominators undermine the original electoral findings. We retain both studies in Appendix A; their disagreement cautions against inferring individual behavior from aggregate electoral contrasts.

enter the program by identifying and registering eligible households. Federal quota revisions during the 2009–2012 mayoral term changed the number of families each municipality could enroll. We use this variation to study the electoral return to program expansion among mayors who sought reelection.

The additional authorization led to more families receiving benefits and greater support for incumbent mayors. A one-standard-deviation quota shock opened 171.50 additional slots per 10,000 residents and induced 54.68 additional beneficiary families per 10,000 residents. Under the identifying assumptions, an expansion of 100 beneficiary families per 10,000 residents raised incumbent vote share by 2.59 percentage points, with Anderson–Rubin 95 percent set $[0.68, 5.67]$. This estimate concerns the enrollment expansion induced by federal quota changes. It does not reveal which voters changed their support.

Local welfare offices provide a plausible setting in which families encounter municipal officials and assign credit for receiving a federal benefit. Enrollment grew both with and without a pre-existing *CRAS*, while initial electoral estimates were larger where an office was present. The contrast weakened after balancing observed municipal characteristics. Comparisons by active outreach and family follow-up motivate the contact interpretation, but detailed tests of reported practices and operating duration do not establish a consistent electoral gradient. Office presence and contact therefore remain possible explanations for the electoral return, rather than separately identified causes.

The broader implication concerns the division of responsibility within welfare programs. Centralizing eligibility and payments leaves local officials involved in families' access to benefits, and those officials can receive an electoral return. Distinguishing credit for registration from material gains, electoral participation, or accompanying services requires additional identifying variation. Our evidence establishes the political relevance of local implementation while leaving that distinction open.

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# Figures and Tables

**Table 1:** *Bolsa Família* Eligibility Thresholds and Benefit Values, 2004–2018

| | 2004 Jan. | 2006 Apr. | 2007 Aug. | 2008 Mar. | 2008 July | 2009 Apr. | 2009 Sept. | 2011 Apr. | 2011 Sept. | 2014 June | 2016 July | 2018 July |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| *Panel A: Monthly income eligibility thresholds (BRL per capita)* | | | | | | | | | | | | |
| Poverty line | 100 | 120 | 120 | 120 | 120 | 137 | 140 | 140 | 140 | 154 | 170 | 178 |
| Extreme-poverty line | 50 | 60 | 60 | 60 | 60 | 69 | 70 | 70 | 70 | 77 | 85 | 89 |
| *Panel B: Monthly benefit values (current BRL) and family caps* | | | | | | | | | | | | |
| Basic benefit (per family) | 50 | 50 | 58 | 58 | 62 | 62 | 68 | 70 | 70 | 77 | 85 | 89 |
| Variable benefit (per member) | 15 | 15 | 18 | 18 | 20 | 20 | 22 | 32 | 32 | 35 | 39 | 41 |
| Variable-benefit cap (number) | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 5 | 5 | 5 | 5 |
| Youth benefit (ages 16–17) | – | – | – | 30 | 30 | 30 | 33 | 38 | 38 | 42 | 46 | 48 |
| Youth-benefit cap (number) | – | – | – | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 |

*Notes:* Values are nominal monthly amounts, not average payments or inflation-adjusted benefits. Columns show selected policy-change dates; unchanged values carry forward through December 2018. The basic benefit applies to extremely poor families. Variable benefits depend on family composition and the displayed caps; youth benefits apply to school-enrolled teenagers aged 16–17. A dash means the youth component had not yet been introduced. The table excludes transitional compensation for families entering from predecessor programs and subnational supplements.

*Timing:* Columns use payment-effect months when benefits change. The poverty thresholds of 140/70, 154/77, and 170/85 took legal effect on July 31, 2009, May 2, 2014, and June 30, 2016, respectively, before the corresponding payment columns. The variable-benefit cap rose by decree in June 2011 and was implemented in September. Payments for nursing mothers and pregnant women began in November and December 2011 within that same cap.

*Gap-filling supplement:* From June 2012, qualifying recipient families with a child aged 0–6 could receive an additional benefit to raise post-transfer per-capita income above the extreme-poverty line. The child-age condition expanded to 0–15 in December 2012 and was removed from March 2013. This supplement is family-specific, not another flat amount: the rule multiplies the gap to the extreme-poverty line plus BRL 0.01 by family size and rounds up to the next higher multiple of BRL 2. These changes left the basic, variable, and youth amounts unchanged until 2014.

*Sources:* Law 10,836/2004; Provisional Measure 411/2007; Decrees 5,749, 6,157, 6,491, 6,824, 6,917, 7,447, 7,494, 7,758, 7,852, 8,232, 8,794, 9,396; Law 12,817/2013; MDS Informe 158 and Informe 285. Each linked source and parameter change is recorded in the companion source audit.

**Table 2:** Construction of the Municipal Estimation Sample

| Restriction | Excluded | Remaining |
|---|---|---|
| Integrated municipal identifiers, 2008 and 2012 | – | 5,568 |
| Observed quota shocks and enrollment measures | 63 | 5,505 |
| Observed population and remaining reference covariates | 0 | 5,505 |
| The 2008 winner runs again in 2012 | 3,140 | 2,365 |
| Observed electoral outcomes and matching fields | 1 | 2,364 |
| More than one candidate in the decisive round | 83 | 2,281 |
| Positive vote share for the tracked mayor | 84 | 2,197 |
| Valid votes and turnout meet election-quality thresholds | 0 | 2,197 |
| No ambiguous candidate-name match | 7 | 2,190 |
| Consistent electoral turnout identities | 0 | 2,190 |
| No multiple elected-status records | 85 | 2,105 |
| Positive population denominator and weight | 0 | 2,105 |
| Both election years satisfy the reference restrictions | 0 | 2,105 |
| Observed municipal transfer measures | 24 | 2,081 |
| Observed mayor characteristics | 3 | 2,078 |
| Remaining common-specification fields | 0 | 2,078 |
| Both years satisfy all estimation requirements | 0 | 2,078 |

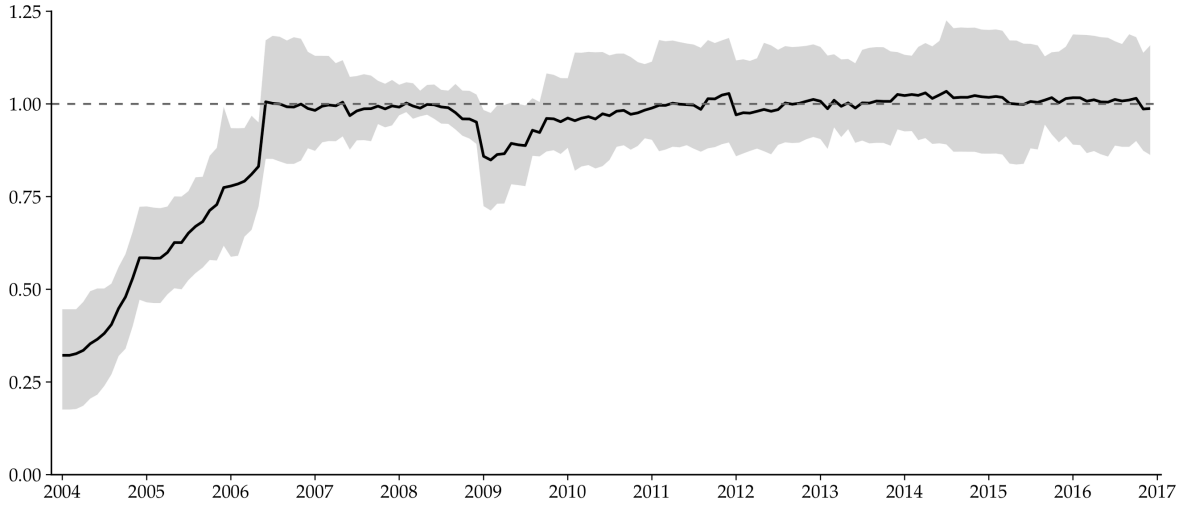
*Notes:* Counts refer to distinct municipal identifiers. Restrictions are cumulative in the displayed order. Every restriction requires both 2008 and 2012 to remain observed. The reporting order separates program availability, rerunning, electoral quality, and remaining covariates. Missingness categories can overlap, so losses depend on the reporting order. The initial identifier universe spans the integrated multi-election database. Complete program measures include the quota instruments and realized enrollment for both analysis windows. Election validity requires positive valid votes, turnout of at least 50 percent, and valid votes of at least 20 percent of turnout. Common-specification requirements include mayor controls, transfer measures, alternative instruments, and mechanism indicators. The final panel contains two observations per municipality. The descriptive and selection-diagnostic universes use separate restrictions.

## **Table 3:** Descriptive Statistics

| **Panel A: All Municipalities** | | | | | | | **Panel B: Regression Sample** | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | N | Min | Mean | Median | Max | SD | | N | Min | Mean | Median | Max | SD |
| Population (2000) | 4,976 | 795.00 | 31,515.59 | 10,703.50 | 10,277,518.00 | 190,735.22 | Population (2000) | 2,078 | 795.00 | 27,991.14 | 10,556.50 | 5,773,130.00 | 146,177.39 |
| Urban Population (2000) (%) | 4,976 | 0.00 | 59.29 | 59.92 | 100.00 | 23.32 | Urban Population (2000) (%) | 2,078 | 0.00 | 58.80 | 59.35 | 100.00 | 23.32 |
| Poor Population (2000) (%) | 4,976 | 5.17 | 64.02 | 66.23 | 99.00 | 20.76 | Poor Population (2000) (%) | 2,078 | 7.66 | 64.09 | 66.80 | 99.00 | 20.99 |
| Income per capita (2000) (R$) | 4,976 | 62.65 | 338.27 | 306.00 | 1,759.76 | 193.29 | Income per capita (2000) (R$) | 2,078 | 69.20 | 337.40 | 300.13 | 1,759.76 | 195.41 |
| Gini (2000) (%) | 4,976 | 30.00 | 54.78 | 55.00 | 82.00 | 6.80 | Gini (2000) (%) | 2,078 | 30.00 | 54.84 | 55.00 | 80.00 | 6.73 |
| Formal Workers (2000) (%) | 4,976 | 1.92 | 36.15 | 34.44 | 86.38 | 18.16 | Formal Workers (2000) (%) | 2,078 | 2.89 | 35.80 | 33.98 | 86.38 | 18.31 |
| Permitted *Bolsa Família* Families (2008) | 4,976 | 24.00 | 2,078.06 | 917.50 | 285,060.00 | 6,980.68 | Permitted *Bolsa Família* Families (2008) | 2,078 | 24.00 | 1,898.41 | 906.00 | 164,913.00 | 5,077.26 |
| Permitted *Bolsa Família* Families (2008) (%) | 4,976 | 0.77 | 8.81 | 8.29 | 67.43 | 4.66 | Permitted *Bolsa Família* Families (2008) (%) | 2,078 | 0.77 | 8.89 | 8.39 | 33.00 | 4.71 |
| Permitted *Bolsa Família* Families (2008) per 100 Families | 4,976 | 2.89 | 34.32 | 30.33 | 113.31 | 18.66 | Permitted *Bolsa Família* Families (2008) per 100 Families | 2,078 | 2.89 | 34.60 | 31.12 | 110.30 | 18.94 |
| Permitted *Bolsa Família* Families (2008) per 10,000 Residents | 4,976 | 77.08 | 880.65 | 829.13 | 6,742.82 | 466.41 | Permitted *Bolsa Família* Families (2008) per 10,000 Residents | 2,078 | 77.08 | 888.88 | 839.23 | 3,299.89 | 470.55 |
| Δ Permitted *Bolsa Família* Families per 100 Families | 4,976 | -36.28 | 5.59 | 3.48 | 103.48 | 11.72 | Δ Permitted *Bolsa Família* Families per 100 Families | 2,078 | -36.28 | 5.71 | 3.57 | 86.01 | 11.74 |
| Δ Permitted *Bolsa Família* Families per 10,000 Residents | 4,976 | -2,841.24 | 116.55 | 91.84 | 2,865.11 | 274.55 | Δ Permitted *Bolsa Família* Families per 10,000 Residents | 2,078 | -1,000.62 | 120.42 | 95.75 | 2,865.11 | 277.26 |
| Realized *Bolsa Família* Families (2008) | 4,976 | 2.42 | 2,045.08 | 914.79 | 204,410.17 | 5,990.93 | Realized *Bolsa Família* Families (2008) | 2,078 | 5.50 | 1,871.27 | 908.38 | 133,322.75 | 4,543.18 |
| Realized *Bolsa Família* Families (2008) (%) | 4,976 | 0.08 | 8.76 | 8.13 | 61.49 | 4.91 | Realized *Bolsa Família* Families (2008) (%) | 2,078 | 0.26 | 8.80 | 8.31 | 29.64 | 4.90 |
| Realized *Bolsa Família* Families (2008) per 100 Families | 4,976 | 0.28 | 34.49 | 30.03 | 105.92 | 20.21 | Realized *Bolsa Família* Families (2008) per 100 Families | 2,078 | 0.93 | 34.58 | 30.84 | 89.54 | 20.26 |
| Realized *Bolsa Família* Families (2008) per 10,000 Residents | 4,976 | 8.48 | 876.40 | 813.07 | 6,149.19 | 491.21 | Realized *Bolsa Família* Families (2008) per 10,000 Residents | 2,078 | 25.81 | 879.87 | 830.99 | 2,963.95 | 490.46 |
| Δ Realized *Bolsa Família* Families per 100 Families | 4,976 | -13.85 | 7.78 | 5.26 | 93.70 | 10.06 | Δ Realized *Bolsa Família* Families per 100 Families | 2,078 | -12.71 | 8.63 | 6.09 | 93.70 | 10.28 |
| Δ Realized *Bolsa Família* Families per 10,000 Residents | 4,976 | -1,234.13 | 180.99 | 139.04 | 2,979.12 | 231.70 | Δ Realized *Bolsa Família* Families per 10,000 Residents | 2,078 | -398.59 | 203.50 | 163.15 | 2,979.12 | 238.38 |
| Candidates (2008) | 4,976 | 2.00 | 2.73 | 2.00 | 12.00 | 1.02 | Candidates (2008) | 2,078 | 2.00 | 2.80 | 3.00 | 12.00 | 1.07 |
| Candidates (2012) | 4,976 | 2.00 | 2.77 | 2.00 | 12.00 | 1.08 | Candidates (2012) | 2,078 | 2.00 | 2.70 | 2.00 | 9.00 | 1.00 |
| Trying Reelection (2012) (%) | 4,976 | 0.00 | 43.75 | 0.00 | 100.00 | 49.61 | Trying Reelection (2012) (%) | 2,078 | 100.00 | 100.00 | 100.00 | 100.00 | 0.00 |
| Winner's Vote Share (2008) (%) | 4,976 | 24.06 | 54.97 | 53.73 | 99.98 | 10.46 | Winner's Vote Share (2008) (%) | 2,078 | 24.06 | 52.92 | 52.68 | 99.16 | 9.43 |
| Incumbent Vote Share (2012) (%) | 2,177 | 0.00 | 48.28 | 49.24 | 100.00 | 16.41 | Incumbent Vote Share (2012) (%) | 2,078 | 0.18 | 49.99 | 49.65 | 100.00 | 14.03 |
| Δ Incumbent Vote Share (pp) | 2,177 | -79.46 | -4.60 | -3.60 | 55.47 | 17.29 | Δ Incumbent Vote Share (pp) | 2,078 | -71.82 | -2.94 | -3.03 | 55.47 | 15.13 |
| Female Mayor (2008) (%) | 4,970 | 0.00 | 9.22 | 0.00 | 100.00 | 28.93 | Female Mayor (2008) (%) | 2,078 | 0.00 | 10.15 | 0.00 | 100.00 | 30.21 |
| High School or More (2008) (%) | 4,970 | 0.00 | 78.85 | 100.00 | 100.00 | 40.84 | High School or More (2008) (%) | 2,078 | 0.00 | 81.14 | 100.00 | 100.00 | 39.13 |
| College Degree (2008) (%) | 4,970 | 0.00 | 45.11 | 0.00 | 100.00 | 49.77 | College Degree (2008) (%) | 2,078 | 0.00 | 46.39 | 0.00 | 100.00 | 49.88 |
| Mayor Age (2008) | 4,970 | 21.00 | 48.16 | 48.00 | 90.00 | 9.52 | Mayor Age (2008) | 2,078 | 21.00 | 46.51 | 46.00 | 81.00 | 9.08 |
| Married (2008) (%) | 4,970 | 0.00 | 78.13 | 100.00 | 100.00 | 41.34 | Married (2008) (%) | 2,078 | 0.00 | 77.05 | 100.00 | 100.00 | 42.06 |
| Public-Sector Occupation (2008) (%) | 4,970 | 0.00 | 34.21 | 0.00 | 100.00 | 47.44 | Public-Sector Occupation (2008) (%) | 2,078 | 0.00 | 15.74 | 0.00 | 100.00 | 36.42 |
| Agricultural Occupation (2008) (%) | 4,970 | 0.00 | 12.03 | 0.00 | 100.00 | 32.54 | Agricultural Occupation (2008) (%) | 2,078 | 0.00 | 14.49 | 0.00 | 100.00 | 35.20 |
| Business Occupation (2008) (%) | 4,970 | 0.00 | 20.24 | 0.00 | 100.00 | 40.18 | Business Occupation (2008) (%) | 2,078 | 0.00 | 26.32 | 0.00 | 100.00 | 44.05 |

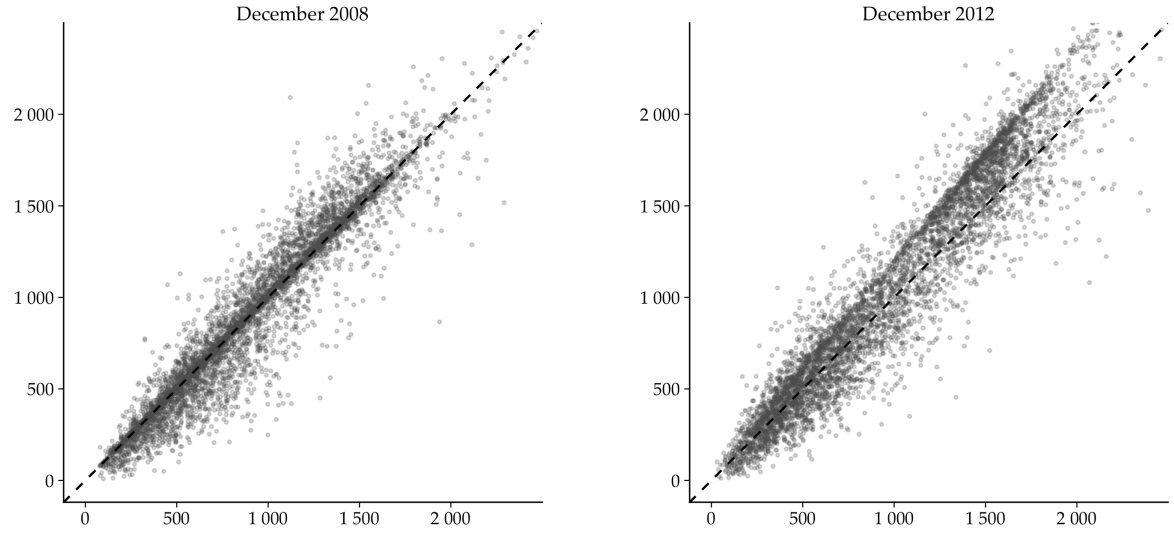
*Notes:* Each observation is one municipality. Panel A contains municipalities with a balanced 2008–2012 election panel that satisfy the common election-quality restrictions; *N* is the number with nonmissing values in each row. Panel B is the exact locked regression sample used in the headline estimates (2,078 municipalities). Permitted families in 2008 are the pre-reform authorized quota (EST 2006); realized families are 12-month averages. Percentage and per-10,000 measures use fixed 2006 population, while per-100-family measures use Census-2000 families. Every Δ is the corresponding 2012 value minus its 2008 value. Candidate counts describe first-round mayoral candidates. Trying reelection indicates that the sitting mayor contested the 2012 election. Vote share tracks the 2008 winner: the 2008 value is that winner's baseline share, and the 2012 value and its change are observed only when the same mayor reran. Mayor characteristics describe the 2008 winner and match the headline controls; age squared is omitted because it is a deterministic transformation of age.

**Figure 1:** Realized Enrollment as a Share of Authorized Quota



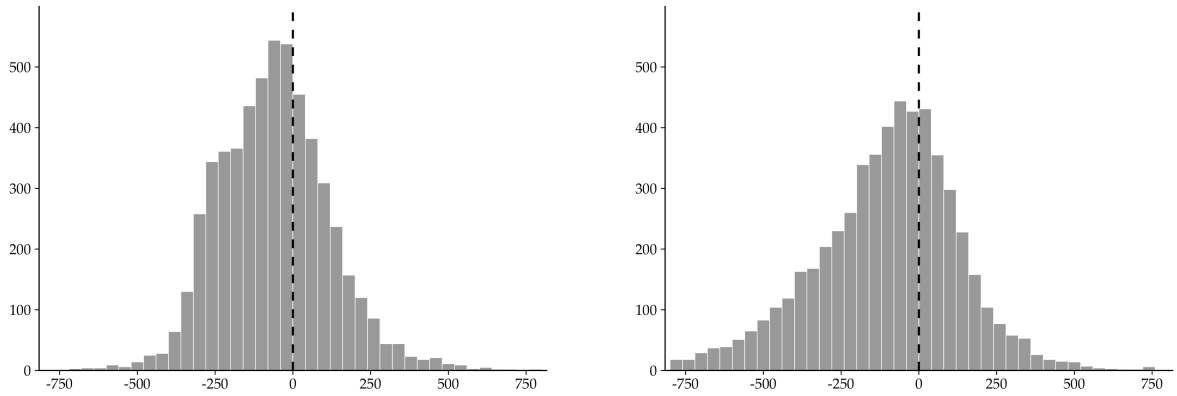

*Note.* For each municipality-month, we compute the ratio of realized *Bolsa Família* beneficiaries to the authorized quota. The solid line plots the quota-weighted cross-municipal mean of these municipality-level ratios. The shaded band spans the weighted 25th to 75th percentile. The dashed horizontal line marks full take-up (ratio = 1). Before 2007, municipalities operated well below their authorized ceilings, and the rapid convergence toward 1 reflects the administrative build-up of the program in its first years. After this initial convergence, realized enrollment generally tracked authorized capacity closely. The 2009 reform temporarily broke that alignment: expanded quotas opened a gap between authorized slots and realized beneficiaries, creating the local implementation margin through which municipalities had to identify, register, and enroll additional eligible families.

**Figure 2:** Authorized Quota vs. Realized Enrollment




*Note.* Each dot is a municipality. The horizontal axis shows the federally authorized quota and the vertical axis shows the 12-month average stock of families enrolled in *Bolsa Família*, both per 10,000 residents using fixed 2006 population. The dashed line marks the 45-degree locus of full take-up (realized = authorized). The panels show December 2008, before the mayoral term, and December 2012, at its endpoint. Both axes share a cap at the 99th percentile of the displayed cross-municipal distributions (approximately 2,400 beneficiary families per 10,000 residents); observations above this threshold are excluded from the frame but not from any analysis.

**Figure 3:** Distribution of Slack Between Authorized Quota and Realized Enrollment




*Note.* Each panel shows slack in December 2012 per 10,000 residents using fixed 2006 population. Slack is the relevant authorized quota minus the 12-month average stock of beneficiary families in 2012. The left panel uses the official 2012 quota; the right panel uses the counterfactual 2009 quota implied by the previous allocation rule. Positive values indicate authorized room not translated into realized enrollment, while negative values indicate realized enrollment above the relevant quota. The dashed vertical line marks zero slack. Both panels retain the common display window from -800 to 800 beneficiary families per 10,000 residents to limit the influence of extreme observations.

**Table 4:** First-Stage Estimates

| | $\Delta$Beneficiaries$_m$ | | |
|---|---|---|---|
| | (1) | (2) | (3) |
| $\Delta$Quota$_m$ ($\pi_1$) | 0.317 | 0.359 | 0.301 |
| | ( 0.063 ) | ( 0.062 ) | ( 0.052 ) |
| Kleibergen–Paap $F$ | 25.43 | 33.63 | 33.30 |
| Incumbent Party | ✓ | ✓ | ✓ |
| *Bolsa Família* Intensity | ✓ | ✓ | ✓ |
| Municipality Characteristics | | ✓ | ✓ |
| Mayor Characteristics | | ✓ | ✓ |
| Benefit Value | | | ✓ |
| *FPM* Transfers | | | ✓ |
| Number of Mayors | 2,078 | 2,078 | 2,078 |
| Observations | 4,156 | 4,156 | 4,156 |

*Notes:* All columns use 2006-population weights and state-clustered standard errors. The instrument and endogenous variable are rates per 10,000 residents using fixed 2006 population and are reported per one analysis-weighted SD. Column (1) includes incumbent-party and baseline program-intensity controls; Column (2) adds municipality and mayor characteristics; Column (3) adds 2008 benefit value and FPM transfers, each interacted with post. Standard errors are in parentheses.

**Table 5:** Reduced-Form Estimates

| | Δ Incumbent's Vote Share | | | Δ Turnout | | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| $\Delta\text{Quota}_m$ ($\rho_1$) | 1.929 | 1.299 | 1.414 | -0.372 | -0.352 | -0.354 |
| | ( 0.604 ) | ( 0.461 ) | ( 0.482 ) | ( 0.192 ) | ( 0.181 ) | ( 0.180 ) |
| Incumbent Party | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| *Bolsa Família* Intensity | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Municipality Characteristics | | ✓ | ✓ | | ✓ | ✓ |
| Mayor Characteristics | | ✓ | ✓ | | ✓ | ✓ |
| Benefit Value | | | ✓ | | | ✓ |
| *FPM* Transfers | | | ✓ | | | ✓ |
| Number of Mayors | 2,078 | 2,078 | 2,078 | 2,078 | 2,078 | 2,078 |
| Observations | 4,156 | 4,156 | 4,156 | 4,156 | 4,156 | 4,156 |

*Notes:* All columns use 2006-population weights and state-clustered standard errors. The instrument and endogenous variable are rates per 10,000 residents using fixed 2006 population and are reported per one analysis-weighted SD. Column (1) includes incumbent-party and baseline program-intensity controls; Column (2) adds municipality and mayor characteristics; Column (3) adds 2008 benefit value and FPM transfers, each interacted with post. Standard errors are in parentheses.

**Table 6:** Two-Stage Least Squares Estimates

| | Δ Incumbent's Vote Share | | | Δ Turnout | | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| $\Delta\widehat{\text{Beneficiaries}}_m$ ($\beta_1$) | 6.082 | 3.619 | 4.690 | -1.173 | -0.981 | -1.173 |
| | ( 2.322 ) | ( 1.569 ) | ( 1.991 ) | ( 0.698 ) | ( 0.532 ) | ( 0.639 ) |
| AR 95% CI | [2.019, 12.834] | [0.876, 7.978] | [1.242, 10.282] | [-3.168, 0.069] | [-2.302, 0.060] | [-2.792, 0.053] |
| Incumbent Party | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| *Bolsa Família* Intensity | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Municipality Characteristics | | ✓ | ✓ | | ✓ | ✓ |
| Mayor Characteristics | | ✓ | ✓ | | ✓ | ✓ |
| Benefit Value | | | ✓ | | | ✓ |
| *FPM* Transfers | | | ✓ | | | ✓ |
| Number of Mayors | 2,078 | 2,078 | 2,078 | 2,078 | 2,078 | 2,078 |
| Observations | 4,156 | 4,156 | 4,156 | 4,156 | 4,156 | 4,156 |

*Notes:* All columns use 2006-population weights and state-clustered standard errors. The instrument and endogenous variable are rates per 10,000 residents using fixed 2006 population and are reported per one analysis-weighted SD. Column (1) includes incumbent-party and baseline program-intensity controls; Column (2) adds municipality and mayor characteristics; Column (3) adds 2008 benefit value and FPM transfers, each interacted with post. Standard errors are in parentheses. The Kleibergen–Paap statistic is the squared state-clustered first-stage t-ratio. Anderson–Rubin 95% confidence sets use quadratic state-clustered inversion with a $t_{G-1}$ critical value and are robust to arbitrary instrument weakness.

**Table 7:** Heterogeneity by Municipality Size

| | Δ Incumbent's Vote Share | | | Δ Turnout | | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| *Panel A: Below Median* | | | | | | |
| $\Delta\text{Quota}_m$ $(\rho_1)$ | -0.147 | -0.057 | 0.033 | -0.167 | -0.172 | -0.223 |
| | ( 0.318 ) | ( 0.289 ) | ( 0.292 ) | ( 0.117 ) | ( 0.103 ) | ( 0.109 ) |
| $\Delta\widehat{\text{Beneficiaries}}_m$ $(\beta_1)$ | -0.303 | -0.114 | 0.073 | -0.345 | -0.347 | -0.493 |
| | ( 0.675 ) | ( 0.585 ) | ( 0.643 ) | ( 0.260 ) | ( 0.216 ) | ( 0.265 ) |
| AR 95% CI | [-1.979, 0.961] | [-1.473, 1.040] | [-1.435, 1.369] | [-0.985, 0.145] | [-0.851, 0.078] | [-1.164, 0.003] |
| *Panel B: Below the 75th Percentile* | | | | | | |
| $\Delta\text{Quota}_m$ $(\rho_1)$ | -0.078 | -0.138 | 0.042 | -0.290 | -0.230 | -0.291 |
| | ( 0.315 ) | ( 0.329 ) | ( 0.291 ) | ( 0.198 ) | ( 0.159 ) | ( 0.156 ) |
| $\Delta\widehat{\text{Beneficiaries}}_m$ $(\beta_1)$ | -0.200 | -0.312 | 0.109 | -0.740 | -0.519 | -0.755 |
| | ( 0.800 ) | ( 0.740 ) | ( 0.756 ) | ( 0.571 ) | ( 0.384 ) | ( 0.448 ) |
| AR 95% CI | [-1.844, 1.662] | [-1.881, 1.275] | [-1.468, 1.790] | [-2.288, 0.267] | [-1.442, 0.208] | [-1.879, 0.075] |
| *Panel C: Above the 75th Percentile* | | | | | | |
| $\Delta\text{Quota}_m$ $(\rho_1)$ | 3.873 | 2.584 | 2.908 | -0.412 | -0.487 | -0.430 |
| | ( 1.113 ) | ( 0.883 ) | ( 0.928 ) | ( 0.203 ) | ( 0.246 ) | ( 0.278 ) |
| $\Delta\widehat{\text{Beneficiaries}}_m$ $(\beta_1)$ | 12.596 | 7.731 | 11.270 | -1.341 | -1.458 | -1.666 |
| | ( 4.061 ) | ( 3.173 ) | ( 4.840 ) | ( 0.775 ) | ( 0.772 ) | ( 1.158 ) |
| AR 95% CI | [5.274, 24.648] | [2.195, 17.645] | [3.368, 27.324] | [-3.703, 0.018] | [-3.592, 0.065] | [-4.931, 0.558] |
| *Panel D: Above the 90th Percentile* | | | | | | |
| $\Delta\text{Quota}_m$ $(\rho_1)$ | 6.790 | 5.301 | 5.424 | -0.346 | -0.537 | -0.448 |
| | ( 1.425 ) | ( 0.947 ) | ( 1.048 ) | ( 0.204 ) | ( 0.312 ) | ( 0.345 ) |
| $\Delta\widehat{\text{Beneficiaries}}_m$ $(\beta_1)$ | 20.717 | 14.167 | 18.257 | -1.056 | -1.434 | -1.509 |
| | ( 7.342 ) | ( 4.990 ) | ( 7.058 ) | ( 0.824 ) | ( 0.989 ) | ( 1.383 ) |
| AR 95% CI | [10.647, 71.349] | [7.342, 42.878] | [8.677, 60.479] | [-6.167, 0.200] | [-5.886, 0.294] | [-8.196, 0.813] |
| Incumbent Party | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| *Bolsa Família* Intensity | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Municipality Characteristics | | ✓ | ✓ | | ✓ | ✓ |
| Mayor Characteristics | | ✓ | ✓ | | ✓ | ✓ |
| Benefit Value | | | ✓ | | | ✓ |
| *FPM* Transfers | | | ✓ | | | ✓ |

*Notes:* Panels use the 2006-population median, 75th percentile, and 90th percentile calculated in the headline sample. The below-median and below-75th panels are nested; the above-90th-percentile panel is a subset of the above-75th panel. Models are estimated separately within each displayed sample. Columns follow the headline control sequence; all other specification choices match the main tables. No mean column is reported.

**Table 8:** Heterogeneity by Subgroup: Frey Sample and Baseline *Bolsa Família* Intensity

| | Frey 2019 | | | Baseline *Bolsa Família* Intensity | | |
|---|---|---|---|---|---|---|
| | Outside Sample | Inside Sample | Δ | Below Median | Above Median | Δ |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| *Panel A: First Stage* | | | | | | |
| $\Delta\text{Quota}_m$ $(\pi_1)$ | 0.213 | 0.366 | -0.153 | 0.255 | 0.370 | -0.115 |
| | ( 0.068 ) | ( 0.055 ) | ( 0.064 ) | ( 0.044 ) | ( 0.055 ) | ( 0.057 ) |
| Conditional First-Stage $F$ | 9.75 | 44.76 | | 33.42 | 45.09 | |
| *Panel B: Reduced Form* | | | | | | |
| $\Delta\text{Quota}_m$ $(\rho_1)$ | 3.017 | -0.045 | 3.062 | 2.603 | -0.150 | 2.753 |
| | ( 0.962 ) | ( 0.270 ) | ( 0.994 ) | ( 1.056 ) | ( 0.222 ) | ( 1.162 ) |
| *Panel C: Two-Stage Least Squares* | | | | | | |
| $\Delta\widehat{\text{Beneficiaries}}_m$ $(\beta_1)$ | 14.198 | -0.122 | 14.320 | 10.192 | -0.405 | 10.598 |
| | ( 7.255 ) | ( 0.736 ) | ( 7.213 ) | ( 4.301 ) | ( 0.612 ) | ( 4.599 ) |
| Conditional AR 95% CI | [3.867, 52.461] | [-1.706, 1.483] | | [1.723, 20.805] | [-1.841, 0.815] | |
| Incumbent Party | ✓ | ✓ | | ✓ | ✓ | |
| *Bolsa Família* Intensity | ✓ | ✓ | | ✓ | ✓ | |
| Municipality Characteristics | ✓ | ✓ | | ✓ | ✓ | |
| Mayor Characteristics | ✓ | ✓ | | ✓ | ✓ | |
| Benefit Value | ✓ | ✓ | | ✓ | ✓ | |
| *FPM* Transfers | ✓ | ✓ | | ✓ | ✓ | |
| Number of Mayors | 2,078 | 2,078 | 2,078 | 2,078 | 2,078 | 2,078 |
| Observations | 4,156 | 4,156 | 4,156 | 4,156 | 4,156 | 4,156 |

*Notes:* Each three-column block uses the full headline sample. The Frey sample comprises municipalities with 2000 population below 30,000 and 2010 HDI below 0.7. Each block is one fully interacted dual-IV regression. The instrument, endogenous variable, and complete control set are allowed to vary across the complementary subgroup indicators; Δ is the first coefficient minus the second. The saturated system reproduces separate subgroup coefficients while retaining one joint model for the displayed comparison. All models use 2006-population weights and state-clustered standard errors. Contrast columns are calculated from the joint model and therefore do not receive control check marks. Conditional first-stage $F$ statistics follow Sanderson and Windmeijer (2016). Group AR sets are plug-in conditional diagnostics that condition on the other group coefficient, not exact joint weak-IV regions.

**Table 9:** Heterogeneity by Subgroup: Northeast States and PT Incumbent

| | Northeast States | | | PT Incumbent | | |
|---|---|---|---|---|---|---|
| | Other Regions | Northeast | Δ | Other Parties | PT | Δ |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| *Panel A: First Stage* | | | | | | |
| $\Delta\text{Quota}_m$ $(\pi_1)$ | 0.390 | 0.495 | -0.105 | 0.280 | 0.365 | -0.085 |
| | ( 0.075 ) | ( 0.031 ) | ( 0.082 ) | ( 0.036 ) | ( 0.125 ) | ( 0.109 ) |
| Conditional First-Stage $F$ | 26.85 | 247.95 | | 70.40 | 9.58 | |
| *Panel B: Reduced Form* | | | | | | |
| $\Delta\text{Quota}_m$ $(\rho_1)$ | 1.059 | 0.502 | 0.556 | 1.814 | -0.501 | 2.315 |
| | ( 0.871 ) | ( 0.409 ) | ( 0.962 ) | ( 0.630 ) | ( 0.982 ) | ( 1.447 ) |
| *Panel C: Two-Stage Least Squares* | | | | | | |
| $\Delta\widehat{\text{Beneficiaries}}_m$ $(\beta_1)$ | 2.712 | 1.015 | 1.697 | 6.490 | -1.374 | 7.864 |
| | ( 2.396 ) | ( 0.801 ) | ( 2.526 ) | ( 2.716 ) | ( 2.678 ) | ( 4.838 ) |
| Conditional AR 95% CI | [-1.789, 9.135] | [-0.727, 2.604] | | [1.666, 13.098] | [-10.589, 13.499] | |
| Incumbent Party | ✓ | ✓ | | ✓ | ✓ | |
| *Bolsa Família* Intensity | ✓ | ✓ | | ✓ | ✓ | |
| Municipality Characteristics | ✓ | ✓ | | ✓ | ✓ | |
| Mayor Characteristics | ✓ | ✓ | | ✓ | ✓ | |
| Benefit Value | ✓ | ✓ | | ✓ | ✓ | |
| *FPM* Transfers | ✓ | ✓ | | ✓ | ✓ | |
| Number of Mayors | 2,078 | 2,078 | 2,078 | 2,078 | 2,078 | 2,078 |
| Observations | 4,156 | 4,156 | 4,156 | 4,156 | 4,156 | 4,156 |

*Notes:* Each three-column block uses the full headline sample. Each block is one fully interacted dual-IV regression. The instrument, endogenous variable, and complete control set are allowed to vary across the complementary subgroup indicators; Δ is the first coefficient minus the second. The saturated system reproduces separate subgroup coefficients while retaining one joint model for the displayed comparison. All models use 2006-population weights and state-clustered standard errors. Contrast columns are calculated from the joint model and therefore do not receive control check marks. Conditional first-stage $F$ statistics follow Sanderson and Windmeijer (2016). Group AR sets are plug-in conditional diagnostics that condition on the other group coefficient, not exact joint weak-IV regions.

**Table 10:** Robustness Checks: Vote Share

| | Δ Incumbent's Vote Share | | | | | | |
|---|---|---|---|---|---|---|---|
| | | Alt. Outcomes | | Alt. Shock | | Alt. Weights | Alt. Scale |
| | Main | Winsorized | Registered Votes | Winsorized | Limited to 2009 | IPW | Per Families |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) |
| *Panel A: First Stage* | | | | | | | |
| $\Delta\text{Quota}_m$ $(\pi_1)$ | 0.301 | 0.301 | 0.301 | 0.277 | 0.236 | 0.302 | 0.281 |
| | ( 0.052 ) | ( 0.052 ) | ( 0.052 ) | ( 0.043 ) | ( 0.034 ) | ( 0.049 ) | ( 0.048 ) |
| Kleibergen–Paap $F$ | 33.30 | 33.30 | 33.30 | 41.26 | 48.07 | 37.68 | 34.88 |
| *Panel B: Reduced Form* | | | | | | | |
| $\Delta\text{Quota}_m$ $(\rho_1)$ | 1.414 | 1.277 | 1.145 | 1.523 | 0.584 | 1.250 | 1.295 |
| | ( 0.482 ) | ( 0.429 ) | ( 0.407 ) | ( 0.478 ) | ( 0.621 ) | ( 0.452 ) | ( 0.507 ) |
| *Panel C: Two-Stage Least Squares* | | | | | | | |
| $\Delta\widehat{\text{Beneficiaries}}_m$ $(\beta_1)$ | 4.690 | 4.234 | 3.797 | 5.500 | 2.474 | 4.144 | 4.604 |
| | ( 1.991 ) | ( 1.773 ) | ( 1.654 ) | ( 2.082 ) | ( 2.570 ) | ( 1.758 ) | ( 2.236 ) |
| AR 95% CI | [1.242, 10.282] | [1.164, 9.209] | [0.914, 8.408] | [1.782, 11.012] | [-3.245, 7.846] | [0.978, 8.809] | [0.772, 10.906] |
| Incumbent Party | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| *Bolsa Família* Intensity | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Municipality Characteristics | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Mayor Characteristics | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Benefit Value | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| *FPM* Transfers | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Number of Mayors | 2,078 | 2,078 | 2,078 | 2,078 | 2,078 | 2,078 | 2,078 |
| Observations | 4,156 | 4,156 | 4,156 | 4,156 | 4,156 | 4,156 | 4,156 |

*Notes:* Column (1) is the preferred valid-vote specification. Columns (2)–(3) winsorize the outcome at the 1st and 99th percentiles or divide incumbent votes by registered voters. Column (4) winsorizes the quota shock. Column (5) replaces the 2012 authorization with the corresponding 2009 authorization and retains the 2008–2012 beneficiary change. Column (6) combines inverse-selection and population weights. Column (7) expresses the instrument and treatment per 100 Census-2000 families. Every column includes the complete control set and state-clustered standard errors. AR sets use cluster-robust quadratic inversion.

**Table 11:** Continuous-Quota and Discrete-Beneficiary Specifications

| | Δ Incumbent's Vote Share | | | Δ Turnout | | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| *Reduced Form* (*common to all panels*) | | | | | | |
| $\Delta\text{Quota}_m$ ($\rho_1$) | 1.929 | 1.299 | 1.414 | -0.372 | -0.352 | -0.354 |
| | ( 0.604 ) | ( 0.461 ) | ( 0.482 ) | ( 0.192 ) | ( 0.181 ) | ( 0.180 ) |
| *Panel A: Continuous* | | | | | | |
| $\Delta\widehat{\text{Beneficiaries}}_m$ ($\beta_1$) | 6.082 | 3.619 | 4.690 | -1.173 | -0.981 | -1.173 |
| | ( 2.322 ) | ( 1.569 ) | ( 1.991 ) | ( 0.698 ) | ( 0.532 ) | ( 0.639 ) |
| AR 95% CI | [2.019, 12.834] | [0.876, 7.978] | [1.242, 10.282] | [-3.168, 0.069] | [-2.302, 0.060] | [-2.792, 0.053] |
| *Panel B: Above Median* | | | | | | |
| $\mathbf{1}\{\Delta\widehat{\text{Beneficiaries}}_m \geq c_D\}$ ($\beta_1$) | 30.129 | 15.579 | 21.951 | -5.812 | -4.223 | -5.488 |
| | ( 14.379 ) | ( 7.838 ) | ( 11.340 ) | ( 3.641 ) | ( 1.953 ) | ( 2.558 ) |
| AR 95% CI | [9.445, 123.527] | [3.430, 44.533] | [5.090, 67.370] | [-25.814, 0.379] | [-9.013, 0.335] | [-12.189, 0.328] |
| *Panel C: Top 25 Percent* | | | | | | |
| $\mathbf{1}\{\Delta\widehat{\text{Beneficiaries}}_m \geq c_D\}$ ($\beta_1$) | 24.045 | 13.966 | 17.647 | -4.638 | -3.786 | -4.412 |
| | ( 8.792 ) | ( 5.342 ) | ( 6.303 ) | ( 2.742 ) | ( 2.129 ) | ( 2.380 ) |
| AR 95% CI | [8.279, 49.109] | [3.826, 28.051] | [5.440, 33.661] | [-12.464, 0.273] | [-9.444, 0.225] | [-10.448, 0.204] |
| *Panel D: Above the 90th Percentile* | | | | | | |
| $\mathbf{1}\{\Delta\widehat{\text{Beneficiaries}}_m \geq c_D\}$ ($\beta_1$) | 29.649 | 18.877 | 22.446 | -5.719 | -5.117 | -5.612 |
| | ( 12.181 ) | ( 7.744 ) | ( 8.693 ) | ( 3.565 ) | ( 3.176 ) | ( 3.431 ) |
| AR 95% CI | [9.474, 69.956] | [4.972, 41.702] | [6.668, 47.813] | [-17.252, 0.322] | [-15.024, 0.273] | [-16.290, 0.228] |
| Incumbent Party | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| *Bolsa Família* Intensity | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Municipality Characteristics | | ✓ | ✓ | | ✓ | ✓ |
| Mayor Characteristics | | ✓ | ✓ | | ✓ | ✓ |
| Benefit Value | | | ✓ | | | ✓ |
| *FPM* Transfers | | | ✓ | | | ✓ |
| Number of Mayors | 2,078 | 2,078 | 2,078 | 2,078 | 2,078 | 2,078 |
| Observations | 4,156 | 4,156 | 4,156 | 4,156 | 4,156 | 4,156 |

*Notes:* The standardized quota shock remains continuous in every panel. Panel A also retains the continuous beneficiary change. Panels B–D instead define the endogenous variable as an indicator that the municipality's 2008–2012 beneficiary change is at or above the sample median, 75th percentile, or 90th percentile, respectively. The reduced form is identical across panels by construction and is shown once above them. The 2SLS coefficient in Panels B–D is the effect of crossing the displayed beneficiary-expansion threshold. Each column follows the headline control sequence. Weights, clustering, and AR inversion otherwise match the main tables.

**Table 12:** Registration Responses in Common Units

| Registration outcome or contrast | Quota effect (1) | 95% CI (2) | Holm p (3) | Mean (4) |
|---|---|---|---|---|
| All registrations | 40.187 | [20.033, 60.342] | 0.003 | 1624.362 |
| Income at or below R$140 | 36.024 | [19.246, 52.802] | <0.001 | 1225.048 |
| Income above R$140 | 4.163 | [-3.302, 11.628] | 0.271 | 399.314 |
| Income at or below R$70 | 30.017 | [16.374, 43.660] | <0.001 | 920.500 |
| Income above R$70 to R$140 | 6.007 | [1.847, 10.168] | 0.006 | 304.548 |
| At most R$140 minus above R$140 | 31.861 | [15.482, 48.240] | <0.001 | 825.734 |
| At most R$70 minus R$70–140 | 24.010 | [12.812, 35.207] | <0.001 | 615.951 |
| R$70–140 minus above R$140 | 1.844 | [-4.630, 8.318] | 0.568 | -94.766 |

*Notes:* All estimates use one full-sample population-weighted SD of the authorized quota shock (171.496 families per 10,000 residents). Weights are 2006 population; municipality/year effects and the complete baseline specification are included. Parentheses: state-clustered SEs; brackets: 95% CRV1 t intervals. Holm p uses 9,999 null-imposed state wild-cluster draws. Registrations are retrospective cohorts surviving in the 2012 snapshot, classified by 2012 income, not verified PBF receipt. All outcomes use families per 10,000 fixed 2006 residents. Three direct contrasts form one multiplicity family; the five response coefficients form a separate family. The overlapping at-most-140 category is not added to its components. Mean is the unweighted two-period outcome or outcome-difference mean. Sample: 2,078 municipalities and 4,156 observations.

**Table 13:** Heterogeneity by CRAS Presence

| | Δ Incumbent's Vote Share | | | Δ Turnout | | |
|---|---|---|---|---|---|---|
| | Without CRAS | With CRAS | Δ | Without CRAS | With CRAS | Δ |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| *Panel A: First Stage* | | | | | | |
| $\Delta\text{Quota}_m$ $(\pi_1)$ | 0.325 | 0.301 | 0.023 | 0.325 | 0.301 | 0.023 |
| | ( 0.040 ) | ( 0.060 ) | ( 0.056 ) | ( 0.040 ) | ( 0.060 ) | ( 0.056 ) |
| Conditional First-Stage $F$ | 66.76 | 25.05 | | 66.76 | 25.05 | |
| *Panel B: Reduced Form* | | | | | | |
| $\Delta\text{Quota}_m$ $(\rho_1)$ | -0.398 | 1.841 | -2.239 | -0.138 | -0.385 | 0.247 |
| | ( 0.634 ) | ( 0.540 ) | ( 0.736 ) | ( 0.073 ) | ( 0.213 ) | ( 0.206 ) |
| *Panel C: Two-Stage Least Squares* | | | | | | |
| $\Delta\widehat{\text{Beneficiaries}}_m$ $(\beta_1)$ | -1.226 | 6.111 | -7.338 | -0.425 | -1.277 | 0.852 |
| | ( 1.935 ) | ( 2.394 ) | ( 2.518 ) | ( 0.220 ) | ( 0.765 ) | ( 0.733 ) |
| Conditional AR 95% CI | [-5.247, 2.991] | [2.123, 13.449] | | [-0.897, 0.039] | [-3.332, 0.176] | |
| Incumbent Party | ✓ | ✓ | | ✓ | ✓ | |
| *Bolsa Família* Intensity | ✓ | ✓ | | ✓ | ✓ | |
| Municipality Characteristics | ✓ | ✓ | | ✓ | ✓ | |
| Mayor Characteristics | ✓ | ✓ | | ✓ | ✓ | |
| Benefit Value | ✓ | ✓ | | ✓ | ✓ | |
| *FPM* Transfers | ✓ | ✓ | | ✓ | ✓ | |
| Number of Mayors | 2,078 | 2,078 | 2,078 | 2,078 | 2,078 | 2,078 |
| Observations | 4,156 | 4,156 | 4,156 | 4,156 | 4,156 | 4,156 |

*Notes:* Each outcome is estimated in one fully interacted dual-IV system allowing the standardized instrument, treatment, and complete control set to vary with 2008 *CRAS* presence and its complement; Δ is the first coefficient minus the second. The saturated system reproduces separate subgroup coefficients while retaining one joint model for the displayed comparison. Models use the headline sample, 2006-population weights, and state-clustered standard errors. Conditional first-stage $F$ statistics follow Sanderson and Windmeijer (2016). AR sets are plug-in conditional diagnostics, not exact joint weak-IV regions.

**Table 14:** Baseline Comparability of Municipalities With and Without CRAS

| CRAS minus no CRAS | Population weights | Overlap weights |
|---|---|---|
| Log population (2006) | 1.128 | 0.000 |
| Urban share (2000) | 0.675 | 0.000 |
| Gini coefficient (2000) | 0.440 | 0.000 |
| Adult illiteracy (2000) | -0.075 | 0.000 |
| Baseline program coverage (2008) | -0.058 | 0.000 |
| Mean family benefit (2008) | 0.318 | 0.000 |
| FPM transfers per resident (2008) | -1.159 | 0.000 |
| Largest absolute state-share gap | 0.152 | 0.000 |
| No CRAS: municipalities | 684 | 684 |
| Effective municipalities | 275.1 | 170.5 |
| CRAS: municipalities | 1394 | 1394 |
| Effective municipalities | 60.9 | 672.1 |

*Notes:* Standardized differences use the fixed population-weighted standard deviation in the full locked sample. The propensity model is a population-weighted logistic regression of 2008 CRAS presence on the seven listed pre-reform covariates and state indicators. Overlap weights multiply 2006 population by one minus the fitted probability for CRAS municipalities and by that probability otherwise. No propensity trimming is applied. Balancing these means is a property of the fitted model, not evidence that unobserved municipal capacity is balanced. Effective municipalities equal the squared sum of weights divided by the sum of squared weights.

**Table 15:** CRAS Quota Responses After Balancing Baseline Characteristics

| | No CRAS | CRAS | Difference |
|---|---|---|---|
| *Enrollment per 10,000 residents* | | | |
| Population weights | 58.878 | 54.639 | -4.239 |
| State-clustered s.e. | (7.206) | (10.916) | (10.212) |
| Wild-cluster $p$ | $< 0.001$ | $< 0.001$ | 0.686 |
| Holm-adjusted $p$ | $< 0.001$ | $< 0.001$ | 0.686 |
| Overlap weights | 59.182 | 67.590 | 8.408 |
| State-clustered s.e. | (7.943) | (11.541) | (8.749) |
| Wild-cluster $p$ | $< 0.001$ | $< 0.001$ | 0.386 |
| Holm-adjusted $p$ | $< 0.001$ | $< 0.001$ | 1.000 |
| *Incumbent vote share (percentage points)* | | | |
| Population weights | -0.398 | 1.841 | 2.239 |
| State-clustered s.e. | (0.634) | (0.540) | (0.736) |
| Wild-cluster $p$ | 0.600 | 0.010 | 0.033 |
| Holm-adjusted $p$ | 0.600 | 0.020 | 0.098 |
| Overlap weights | -0.400 | 0.155 | 0.555 |
| State-clustered s.e. | (0.656) | (0.357) | (0.689) |
| Wild-cluster $p$ | 0.579 | 0.670 | 0.458 |
| Holm-adjusted $p$ | 0.579 | 0.670 | 1.000 |
| *Turnout (percentage points)* | | | |
| Population weights | -0.138 | -0.385 | -0.247 |
| State-clustered s.e. | (0.073) | (0.213) | (0.206) |
| Wild-cluster $p$ | 0.086 | 0.043 | 0.298 |
| Holm-adjusted $p$ | 0.172 | 0.043 | 0.597 |
| Overlap weights | -0.160 | -0.132 | 0.027 |
| State-clustered s.e. | (0.078) | (0.124) | (0.117) |
| Wild-cluster $p$ | 0.062 | 0.333 | 0.836 |
| Holm-adjusted $p$ | 0.124 | 0.666 | 1.000 |

*Notes:* Coefficients are reduced forms per one fixed headline-sample weighted standard deviation of the quota shock; the enrollment response is a first stage in family-count units. The complete baseline specification, party-by-post effects and post indicator are fully interacted with 2008 CRAS status. All models retain the same 2,078 municipalities, 4,156 observations and 26 state clusters. The companion balance table defines overlap weights. State-clustered standard errors and 9,999-draw null-imposed Rademacher wild tests condition on the fitted weights; they do not incorporate propensity-estimation uncertainty. Holm adjustments cover the three outcomes within each weighting scheme and coefficient column. This sensitivity changes the population receiving greatest weight and controls only observed composition; it does not identify the causal effect of opening a CRAS or establish mediation.

**Table 16:** Heterogeneity by CRAS Intensity: Active Outreach

| | Δ Incumbent's Vote Share | | | Δ Turnout | | |
|---|---|---|---|---|---|---|
| | Without Active Outreach | With Active Outreach | Δ | Without Active Outreach | With Active Outreach | Δ |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| *Panel A: First Stage* | | | | | | |
| $\Delta\text{Quota}_m$ $(\pi_1)$ | 0.436 | 0.276 | 0.160 | 0.436 | 0.276 | 0.160 |
| | ( 0.108 ) | ( 0.042 ) | ( 0.084 ) | ( 0.108 ) | ( 0.042 ) | ( 0.084 ) |
| Conditional First-Stage $F$ | 16.23 | 43.13 | | 16.23 | 43.13 | |
| *Panel B: Reduced Form* | | | | | | |
| $\Delta\text{Quota}_m$ $(\rho_1)$ | 0.757 | 2.004 | -1.247 | -0.234 | -0.394 | 0.160 |
| | ( 0.742 ) | ( 0.646 ) | ( 0.897 ) | ( 0.167 ) | ( 0.243 ) | ( 0.155 ) |
| *Panel C: Two-Stage Least Squares* | | | | | | |
| $\Delta\widehat{\text{Beneficiaries}}_m$ $(\beta_1)$ | 1.738 | 7.273 | -5.535 | -0.538 | -1.430 | 0.892 |
| | ( 1.841 ) | ( 2.572 ) | ( 2.702 ) | ( 0.437 ) | ( 0.894 ) | ( 0.647 ) |
| Conditional AR 95% CI | [-1.697, 7.407] | [2.401, 13.661] | | [-1.964, 0.231] | [-3.487, 0.397] | |
| Incumbent Party | ✓ | ✓ | | ✓ | ✓ | |
| *Bolsa Família* Intensity | ✓ | ✓ | | ✓ | ✓ | |
| Municipality Characteristics | ✓ | ✓ | | ✓ | ✓ | |
| Mayor Characteristics | ✓ | ✓ | | ✓ | ✓ | |
| Benefit Value | ✓ | ✓ | | ✓ | ✓ | |
| *FPM* Transfers | ✓ | ✓ | | ✓ | ✓ | |
| Number of Mayors | 1,394 | 1,394 | 1,394 | 1,394 | 1,394 | 1,394 |
| Observations | 2,788 | 2,788 | 2,788 | 2,788 | 2,788 | 2,788 |

*Notes:* Each outcome is estimated within 2008 *CRAS* municipalities in one fully interacted dual-IV system comparing units with and without active outreach; Δ is Without Active Outreach minus With Active Outreach. The instrument, treatment, and complete control set can vary across the two groups. Models use 2006-population weights and state-clustered standard errors. Conditional first-stage $F$ statistics follow Sanderson and Windmeijer (2016). AR sets are plug-in conditional diagnostics, not exact joint weak-IV regions. Groups are predetermined scope conditions, not causal effects of adopting outreach.

**Table 17:** Predetermined CRAS Contact and Quota Responses

*A. Predetermined contact measures and support*

| Measure | Mean | SD | Range | Municipalities | Missing |
|---|---|---|---|---|---|
| General outreach frequency | 0.678 | 0.200 | [0.000, 1.000] | 1394 | 0 |
| Cadastro Unico referrals | 0.981 | 0.119 | [0.000, 1.000] | 1394 | 0 |
| Prioritized family follow-up | 0.730 | 0.182 | [0.000, 1.000] | 1394 | 0 |
| PBF conditionality outreach | 0.596 | 0.263 | [0.000, 1.000] | 1390 | 4 |
| Share of frequently outreaching offices | 0.841 | 0.327 | [0.000, 1.000] | 1394 | 0 |

*B. Primary general-outreach frequency*

| Outcome | Quota at mean (1) | Contact gradient (2) | 95% CI (3) | Holm p (4) | Mean (5) |
|---|---|---|---|---|---|
| Registrations | 40.161<br>(10.513) | 2.203<br>(5.040) | [-8.177, 12.583] | 1.000 | 1731.691 |
| Beneficiary families | 55.191<br>(10.266) | 4.035<br>(4.672) | [-5.587, 13.656] | 1.000 | 1080.183 |
| Incumbent vote share | 1.811<br>(0.551) | 0.394<br>(0.527) | [-0.691, 1.479] | 1.000 | 51.362 |
| Turnout | -0.430<br>(0.221) | 0.145<br>(0.125) | [-0.113, 0.403] | 1.000 | 86.175 |

*Notes:* Quota coefficients use the fixed full-sample weighted SD (171.496 families per 10,000 residents). Contact gradients refer to one fixed population-weighted SD of the 2008 municipal practice index, among baseline CRAS municipalities with observed responses. Office scores are averaged equally within municipalities. Municipality/year effects and all headline baseline controls, party trends and the post indicator interact with the continuous index. Weights are 2006 population. Parentheses contain state-clustered SEs; confidence intervals use CRV1 t critical values. Wild tests use 9,999 null-imposed Rademacher draws clustered by state. All outcomes are changes identified from 2008/2012 levels: registrations and beneficiary families per 10,000 residents; vote share and turnout in percentage points. Registration cohorts and income/residence are retrospective in the 2012 snapshot. Mean is the unweighted two-period outcome mean. These gradients are observational moderation, not effects of changing a practice. The primary Holm family contains four outcome gradients. Panel A means/SDs are population weighted; ranges and missingness are descriptive. Frequency scores are never=0, rarely=1/3, frequently=2/3, very frequently=1. The conditionality item retains invalid answers as missing. Main sample: 1394 municipalities, 2788 observations.

**Table 18:** Secondary CRAS Contact Practices and Quota Responses

| Outcome | Contact gradient (1) | 95% CI (2) | Wild p (3) | Holm p (4) | Mean (5) |
|---|---|---|---|---|---|
| *Cadastro Unico referrals* | | | | | |
| Registrations | 1.720 (2.132) | [-2.672, 6.111] | 0.467 | 1.000 | 1731.691 |
| Beneficiary families | -3.330 (1.362) | [-6.135, -0.526] | 0.046 | 0.371 | 1080.183 |
| Incumbent vote share | 0.197 (0.390) | [-0.606, 1.000] | 0.745 | 1.000 | 51.362 |
| Turnout | -0.155 (0.044) | [-0.246, -0.064] | 0.072 | 0.507 | 86.175 |
| *Prioritized family follow-up* | | | | | |
| Registrations | 4.960 (5.647) | [-6.670, 16.589] | 0.474 | 1.000 | 1731.691 |
| Beneficiary families | -4.301 (4.731) | [-14.044, 5.441] | 0.390 | 1.000 | 1080.183 |
| Incumbent vote share | -0.092 (0.421) | [-0.959, 0.775] | 0.842 | 1.000 | 51.362 |
| Turnout | 0.043 (0.103) | [-0.169, 0.254] | 0.745 | 1.000 | 86.175 |
| *PBF conditionality outreach* | | | | | |
| Registrations | 11.083 (7.832) | [-5.049, 27.214] | 0.297 | 0.891 | 1731.222 |
| Beneficiary families | 9.107 (4.158) | [0.543, 17.671] | 0.091 | 0.365 | 1079.521 |
| Incumbent vote share | -0.386 (0.496) | [-1.409, 0.636] | 0.494 | 0.891 | 51.385 |
| Turnout | 0.121 (0.112) | [-0.110, 0.353] | 0.400 | 0.891 | 86.185 |

*Notes:* Quota coefficients use the fixed full-sample weighted SD (171.496 families per 10,000 residents). Contact gradients refer to one fixed population-weighted SD of the 2008 municipal practice index, among baseline CRAS municipalities with observed responses. Office scores are averaged equally within municipalities. Municipality/year effects and all headline baseline controls, party trends and the post indicator interact with the continuous index. Weights are 2006 population. Parentheses contain state-clustered SEs; confidence intervals use CRV1 t critical values. Wild tests use 9,999 null-imposed Rademacher draws clustered by state. All outcomes are changes identified from 2008/2012 levels: registrations and beneficiary families per 10,000 residents; vote share and turnout in percentage points. Registration cohorts and income/residence are retrospective in the 2012 snapshot. Mean is the unweighted two-period outcome mean. These gradients are observational moderation, not effects of changing a practice. Referral/follow-up share one eight-contrast Holm family. Conditionality is exploratory with a separate four-contrast family. Referral indicates directing families to the Cadastro Unico agency, not completed enrollment. Follow-up uses lists to prioritize transfer-recipient families. Samples and index scaling are reported in the companion support panel and complete CSV.

**Table 19:** Sensitivity of the General-Outreach Contact Gradient

| Outcome | Contact gradient (1) | 95% CI (2) | Wild p (3) | Holm p (4) | Mean (5) |
|---|---|---|---|---|---|
| *Share of frequently outreaching offices* | | | | | |
| Registrations | 5.064 (5.435) | [-6.129, 16.258] | 0.369 | 1.000 | 1731.691 |
| Beneficiary families | 10.340 (5.580) | [-1.152, 21.831] | 0.140 | 0.981 | 1080.183 |
| Incumbent vote share | 0.294 (0.479) | [-0.692, 1.280] | 0.565 | 1.000 | 51.362 |
| Turnout | 0.172 (0.111) | [-0.057, 0.402] | 0.232 | 1.000 | 86.175 |
| *Quota-by-baseline adjustment* | | | | | |
| Registrations | 5.192 (5.497) | [-6.128, 16.513] | 0.329 | 1.000 | 1731.691 |
| Beneficiary families | 7.022 (3.805) | [-0.813, 14.858] | 0.106 | 0.846 | 1080.183 |
| Incumbent vote share | 0.351 (0.581) | [-0.846, 1.548] | 0.592 | 1.000 | 51.362 |
| Turnout | 0.137 (0.133) | [-0.136, 0.410] | 0.430 | 1.000 | 86.175 |

*Notes:* Quota coefficients use the fixed full-sample weighted SD (171.496 families per 10,000 residents). Contact gradients refer to one fixed population-weighted SD of the 2008 municipal practice index, among baseline CRAS municipalities with observed responses. Office scores are averaged equally within municipalities. Municipality/year effects and all headline baseline controls, party trends and the post indicator interact with the continuous index. Weights are 2006 population. Parentheses contain state-clustered SEs; confidence intervals use CRV1 t critical values. Wild tests use 9,999 null-imposed Rademacher draws clustered by state. All outcomes are changes identified from 2008/2012 levels: registrations and beneficiary families per 10,000 residents; vote share and turnout in percentage points. Registration cohorts and income/residence are retrospective in the 2012 snapshot. Mean is the unweighted two-period outcome mean. These gradients are observational moderation, not effects of changing a practice. The eight sensitivity contrasts share one Holm family. The first uses the share of offices reporting frequent/very frequent use, avoiding equally spaced ordinal categories. The second retains the primary score and adds quota interactions with baseline population, urbanization, inequality, illiteracy, program coverage, benefit value, FPM and state indicators. This permits observed baseline characteristics to predict quota-response slopes; it does not balance contact practices or remove unobserved confounding.

**Table 20:** Heterogeneity by CRAS Intensity: *Proteção e Atendimento Integral à Família* (PAIF)

| | Δ Incumbent's Vote Share | | | Δ Turnout | | |
|---|---|---|---|---|---|---|
| | Without PAIF | With PAIF | Δ | Without PAIF | With PAIF | Δ |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| *Panel A: First Stage* | | | | | | |
| $\Delta\text{Quota}_m$ $(\pi_1)$ | 0.372 | 0.292 | 0.079 | 0.372 | 0.292 | 0.079 |
| | ( 0.130 ) | ( 0.047 ) | ( 0.113 ) | ( 0.130 ) | ( 0.047 ) | ( 0.113 ) |
| Conditional First-Stage $F$ | 8.18 | 38.04 | | 8.18 | 38.04 | |
| *Panel B: Reduced Form* | | | | | | |
| $\Delta\text{Quota}_m$ $(\rho_1)$ | 0.653 | 2.251 | -1.598 | -0.442 | -0.405 | -0.037 |
| | ( 0.570 ) | ( 0.791 ) | ( 1.004 ) | ( 0.279 ) | ( 0.204 ) | ( 0.162 ) |
| *Panel C: Two-Stage Least Squares* | | | | | | |
| $\Delta\widehat{\text{Beneficiaries}}_m$ $(\beta_1)$ | 1.758 | 7.697 | -5.938 | -1.189 | -1.385 | 0.196 |
| | ( 1.695 ) | ( 3.103 ) | ( 3.421 ) | ( 0.933 ) | ( 0.738 ) | ( 0.614 ) |
| Conditional AR 95% CI | [-1.501, 9.525] | [2.010, 15.783] | | [-6.261, 0.324] | [-3.195, 0.050] | |
| Incumbent Party | ✓ | ✓ | | ✓ | ✓ | |
| *Bolsa Família* Intensity | ✓ | ✓ | | ✓ | ✓ | |
| Municipality Characteristics | ✓ | ✓ | | ✓ | ✓ | |
| Mayor Characteristics | ✓ | ✓ | | ✓ | ✓ | |
| Benefit Value | ✓ | ✓ | | ✓ | ✓ | |
| *FPM* Transfers | ✓ | ✓ | | ✓ | ✓ | |
| Number of Mayors | 1,394 | 1,394 | 1,394 | 1,394 | 1,394 | 1,394 |
| Observations | 2,788 | 2,788 | 2,788 | 2,788 | 2,788 | 2,788 |

*Notes:* Each outcome is estimated within 2008 *CRAS* municipalities in one fully interacted dual-IV system comparing units with and without PAIF-tracked *Bolsa Família* families; Δ is Without PAIF minus With PAIF. The instrument, treatment, and complete control set can vary across the two groups. Models use 2006-population weights and state-clustered standard errors. Conditional first-stage $F$ statistics follow Sanderson and Windmeijer (2016). AR sets are plug-in conditional diagnostics, not exact joint weak-IV regions. Groups are predetermined scope conditions, not causal effects of adopting PAIF services.

**Table 21:** Heterogeneity by CRAS Intensity: *Proteção e Atendimento Integral à Família* (PAIF) Volume

| | Δ Incumbent's Vote Share | | | Δ Turnout | | |
|---|---|---|---|---|---|---|
| | Below Median | Above Median | Δ | Below Median | Above Median | Δ |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| *Panel A: First Stage* | | | | | | |
| $\Delta\text{Quota}_m$ ($\pi_1$) | 0.328 | 0.285 | 0.043 | 0.328 | 0.285 | 0.043 |
| | ( 0.067 ) | ( 0.046 ) | ( 0.075 ) | ( 0.067 ) | ( 0.046 ) | ( 0.075 ) |
| Conditional First-Stage $F$ | 24.13 | 38.25 | | 24.13 | 38.25 | |
| *Panel B: Reduced Form* | | | | | | |
| $\Delta\text{Quota}_m$ ($\rho_1$) | -0.495 | 5.361 | -5.856 | -0.356 | -0.384 | 0.028 |
| | ( 0.958 ) | ( 1.407 ) | ( 2.026 ) | ( 0.350 ) | ( 0.137 ) | ( 0.343 ) |
| *Panel C: Two-Stage Least Squares* | | | | | | |
| $\Delta\widehat{\text{Beneficiaries}}_m$ ($\beta_1$) | -1.509 | 18.778 | -20.287 | -1.085 | -1.344 | 0.259 |
| | ( 2.871 ) | ( 6.060 ) | ( 7.971 ) | ( 1.045 ) | ( 0.611 ) | ( 1.141 ) |
| Conditional AR 95% CI | [-7.706, 5.337] | [8.009, 35.047] | | [-3.462, 1.278] | [-3.059, -0.306] | |
| Incumbent Party | ✓ | ✓ | | ✓ | ✓ | |
| *Bolsa Família* Intensity | ✓ | ✓ | | ✓ | ✓ | |
| Municipality Characteristics | ✓ | ✓ | | ✓ | ✓ | |
| Mayor Characteristics | ✓ | ✓ | | ✓ | ✓ | |
| Benefit Value | ✓ | ✓ | | ✓ | ✓ | |
| *FPM* Transfers | ✓ | ✓ | | ✓ | ✓ | |
| Number of Mayors | 962 | 962 | 962 | 962 | 962 | 962 |
| Observations | 1,924 | 1,924 | 1,924 | 1,924 | 1,924 | 1,924 |

*Notes:* The sample contains 2008 *CRAS* municipalities reporting a positive number of *Bolsa Família* families followed through PAIF. Low and high volume are defined relative to the median positive family count in the headline sample; Δ is Below Median minus Above Median. The fully interacted dual-IV system allows the instrument, treatment, and complete control set to vary across the two groups. Models use the headline weights and state clustering. Conditional first-stage $F$ statistics follow Sanderson and Windmeijer (2016).

**Table 22:** Predetermined CRAS Operating Duration and Quota Responses

| | Presence / office density | Years of operation |
|---|---|---|
| Enrollment / 10,000 | 8.014 [-35.125, 51.154]<br>$p$: 0.840; $p_H$: 1.000 | -3.176 [-14.859, 8.508]<br>$p$: 0.761; $p_H$: 1.000 |
| With baseline office density | 21.235 [4.973, 37.497]<br>$p$: 0.020; $p_H$: 0.200 | -1.217 [-12.191, 9.757]<br>$p$: 0.886; $p_H$: 1.000 |
| Exclude pre-2003 opening dates | | -7.872 [-19.755, 4.010]<br>$p$: 0.342; $p_H$: 1.000 |
| Incumbent votes (pp) | -0.318 [-2.704, 2.069]<br>$p$: 0.808; $p_H$: 1.000 | 0.788 [-0.049, 1.626]<br>$p$: 0.106; $p_H$: 0.952 |
| With baseline office density | -0.893 [-2.155, 0.370]<br>$p$: 0.201; $p_H$: 1.000 | 0.705 [-0.173, 1.584]<br>$p$: 0.172; $p_H$: 1.000 |
| Exclude pre-2003 opening dates | | 0.635 [-0.197, 1.468]<br>$p$: 0.156; $p_H$: 1.000 |
| Registrations / 10,000 | 2.673 [-52.425, 57.771]<br>$p$: 0.940; $p_H$: 1.000 | -3.502 [-15.969, 8.966]<br>$p$: 0.678; $p_H$: 1.000 |
| Turnout (pp) | -0.308 [-0.779, 0.164]<br>$p$: 0.239; $p_H$: 0.956 | 0.015 [-0.093, 0.122]<br>$p$: 0.790; $p_H$: 1.000 |

*Notes:* Cells give quota-response estimates and state-clustered 95% intervals, followed by 9,999-draw wild $p$ and Holm-adjusted $p_H$. Models use the fixed headline quota standard deviation, municipality/year effects, full baseline controls and population weights unless noted. Group models fully interact all nuisance terms. Receipt/votes and registration/turnout define separate primary/secondary families; slopes and contrasts are adjusted separately. Groups with fewer than ten states or 25 municipalities retain estimates but no inferential interval or bootstrap test. Registrations are retrospective 2009–2012 first registrations observed in the 2012 snapshot. Age uses the earliest fully observed opening date reported in 2008, evaluated at December 2008. Presence is the conditional CRAS contrast at zero duration. In the density row, column one reports the quota interaction with 2008 office density per 10,000 residents; column two reports the age gradient conditional on density. Pre-2003 dates may refer to predecessor facilities and are retained in the main estimate, with a prescribed exclusion sensitivity. The primary continuous family contains ten tests, including these sensitivities. Age is not randomly assigned.

**Table 23:** Quota Responses by CRAS Age Before the Reform

| | No CRAS | Opened 2007–08 | Opened by 2006 | Recent minus none | Old minus recent |
|---|---|---|---|---|---|
| Enrollment / 10,000 | 58.878 [43.966, 73.789]<br>$p$: $< 0.001$; $p_H$: $< 0.001$ | 74.011 [37.960, 110.063]<br>$p$: $< 0.001$; $p_H$: 0.002 | 48.874 [32.929, 64.819]<br>$p$: $< 0.001$; $p_H$: $< 0.001$ | 15.134 [-15.581, 45.848]<br>$p$: 0.534; $p_H$: 0.534 | -25.138 [-54.679, 4.404]<br>$p$: 0.203; $p_H$: 0.405 |
| Incumbent votes (pp) | -0.398 [-1.711, 0.915]<br>$p$: 0.600; $p_H$: 0.600 | 1.002 [-0.115, 2.118]<br>$p$: 0.078; $p_H$: 0.157 | 2.446 [0.985, 3.907]<br>$p$: 0.008; $p_H$: 0.023 | 1.400 [0.113, 2.686]<br>$p$: 0.054; $p_H$: 0.214 | 1.444 [-0.155, 3.043]<br>$p$: 0.077; $p_H$: 0.231 |
| Registrations / 10,000 | 45.367 [23.237, 67.498]<br>$p$: 0.001; $p_H$: 0.006 | 63.177 [25.421, 100.933]<br>$p$: 0.003; $p_H$: 0.017 | 30.121 [9.037, 51.205]<br>$p$: 0.008; $p_H$: 0.025 | 17.810 [-23.331, 58.950]<br>$p$: 0.617; $p_H$: 1.000 | -33.056 [-65.733, -0.380]<br>$p$: 0.115; $p_H$: 0.460 |
| Turnout (pp) | -0.138 [-0.289, 0.013]<br>$p$: 0.086; $p_H$: 0.171 | -0.427 [-0.774, -0.081]<br>$p$: 0.005; $p_H$: 0.020 | -0.373 [-0.882, 0.136]<br>$p$: 0.177; $p_H$: 0.177 | -0.289 [-0.647, 0.068]<br>$p$: 0.122; $p_H$: 0.460 | 0.054 [-0.300, 0.408]<br>$p$: 0.773; $p_H$: 1.000 |
| Municipalities / states | 684 / 22 | 563 / 25 | 831 / 26 | | |
| Effective municipalities | 275.1 | 189.1 | 39.1 | | |

*Notes:* Cells give quota-response estimates and state-clustered 95% intervals, followed by 9,999-draw wild $p$ and Holm-adjusted $p_H$. Models use the fixed headline quota standard deviation, municipality/year effects, full baseline controls and population weights unless noted. Group models fully interact all nuisance terms. Receipt/votes and registration/turnout define separate primary/secondary families; slopes and contrasts are adjusted separately. Groups with fewer than ten states or 25 municipalities retain estimates but no inferential interval or bootstrap test. Registrations are retrospective 2009–2012 first registrations observed in the 2012 snapshot. Canonical baseline CRAS status is preserved; old/recent classifications use 2008-reported dates only. Unknown opening years are excluded and audited. Two adjacent contrasts across two primary outcomes form a four-test family, with a separate four-test secondary family. No causal CRAS effect is identified.

**Table 24:** Older Versus Recent CRAS Offices After Baseline Balancing

| | Opened 2007–08 | Opened by 2006 | Old minus recent |
|---|---|---|---|
| Enrollment / 10,000: population | 74.011 [37.963, 110.059]<br>$p$: $< 0.001$; $p_H$: $< 0.001$ | 48.874 [32.930, 64.817]<br>$p$: $< 0.001$; $p_H$: $< 0.001$ | -25.138 [-54.676, 4.401]<br>$p$: 0.204; $p_H$: 0.612 |
| Enrollment / 10,000: overlap | 71.569 [34.246, 108.892]<br>$p$: $< 0.001$; $p_H$: 0.004 | 59.452 [44.424, 74.481]<br>$p$: $< 0.001$; $p_H$: $< 0.001$ | -12.117 [-41.383, 17.149]<br>$p$: 0.669; $p_H$: 1.000 |
| Incumbent votes (pp): population | 1.002 [-0.115, 2.118]<br>$p$: 0.079; $p_H$: 0.236 | 2.446 [0.985, 3.906]<br>$p$: 0.009; $p_H$: 0.036 | 1.444 [-0.154, 3.043]<br>$p$: 0.078; $p_H$: 0.313 |
| Incumbent votes (pp): overlap | 1.017 [-0.238, 2.272]<br>$p$: 0.125; $p_H$: 0.249 | 0.581 [-0.583, 1.746]<br>$p$: 0.365; $p_H$: 0.365 | -0.436 [-2.036, 1.164]<br>$p$: 0.594; $p_H$: 1.000 |
| Population: effective municipalities | 189.1 | 39.1 | |
| Overlap: effective municipalities | 121.9 | 388.3 | |

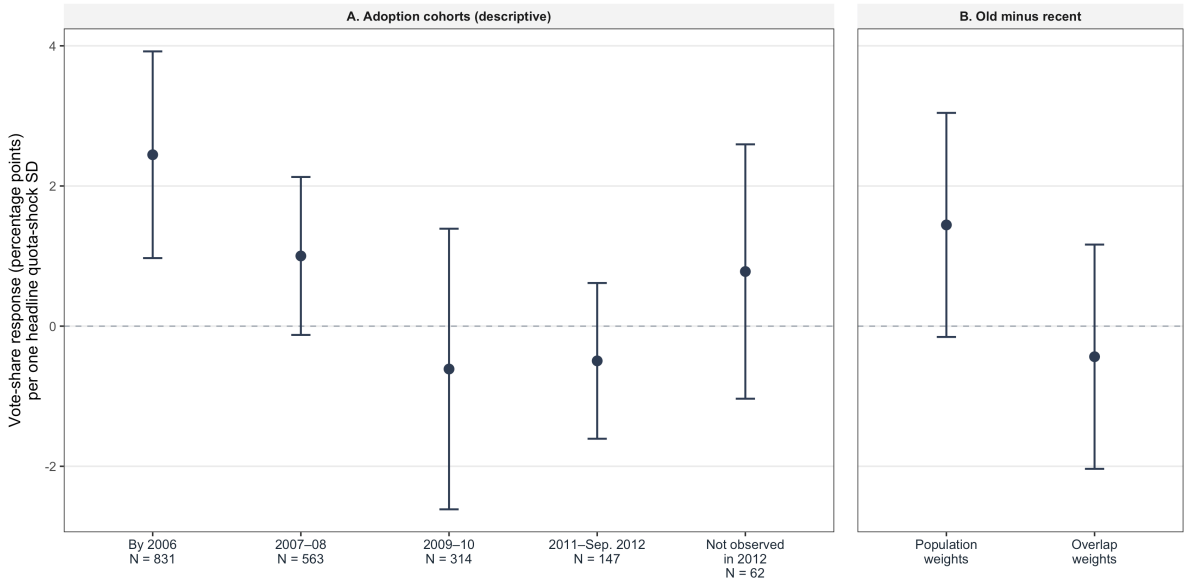
*Notes:* Cells give quota-response estimates and state-clustered 95% intervals, followed by 9,999-draw wild $p$ and Holm-adjusted $p_H$. Models use the fixed headline quota standard deviation, municipality/year effects, full baseline controls and population weights unless noted. Group models fully interact all nuisance terms. Receipt/votes and registration/turnout define separate primary/secondary families; slopes and contrasts are adjusted separately. Groups with fewer than ten states or 25 municipalities retain estimates but no inferential interval or bootstrap test. Registrations are retrospective 2009–2012 first registrations observed in the 2012 snapshot. Pairwise overlap weights compare old and recent baseline CRAS municipalities; they do not balance the no-CRAS group. The population-weighted logit includes baseline population, urbanization, Gini, illiteracy, program coverage, mean benefits, FPM and state indicators. Weights multiply population by the probability of the opposite group. No trimming is applied. Modeled means balance by construction; unobserved differences may remain. Inference conditions on fitted weights. The four old-minus-recent contrasts span both primary outcomes and weighting schemes.

**Table 25:** Descriptive Quota Responses Across CRAS Adoption Cohorts

| | Enrollment / 10,000 | Incumbent votes (pp) | Registrations / 10,000 | Turnout (pp) |
|---|---|---|---|---|
| Baseline: by 2006 | 48.874 [32.774, 64.973] | 2.446 [0.971, 3.921] | 30.121 [8.833, 51.409] | -0.373 [-0.887, 0.141] |
| $N$=831; states=26; eff.$N$=39.1 | $p$: $< 0.001$; $p_H$: $< 0.001$ | $p$: 0.009; $p_H$: 0.053 | $p$: 0.007; $p_H$: 0.051 | $p$: 0.173; $p_H$: 0.692 |
| Baseline: 2007–08 | 74.011 [37.611, 110.411] | 1.002 [-0.126, 2.129] | 63.177 [25.055, 101.298] | -0.427 [-0.778, -0.077] |
| $N$=563; states=25; eff.$N$=189.1 | $p$: $< 0.001$; $p_H$: $< 0.001$ | $p$: 0.081; $p_H$: 0.323 | $p$: 0.002; $p_H$: 0.021 | $p$: 0.004; $p_H$: 0.035 |
| Opened 2009–10 | 62.746 [43.907, 81.584] | -0.612 [-2.614, 1.390] | 46.194 [26.640, 65.748] | -0.317 [-0.549, -0.085] |
| $N$=314; states=20; eff.$N$=124.3 | $p$: $< 0.001$; $p_H$: 0.002 | $p$: 0.600; $p_H$: 0.893 | $p$: 0.006; $p_H$: 0.048 | $p$: 0.028; $p_H$: 0.154 |
| Opened 2011–Sep. 2012 | 78.589 [56.046, 101.132] | -0.496 [-1.607, 0.615] | 96.626 [42.918, 150.334] | 0.057 [-0.216, 0.330] |
| $N$=147; states=14; eff.$N$=70.0 | $p$: $< 0.001$; $p_H$: 0.006 | $p$: 0.298; $p_H$: 0.893 | $p$: 0.026; $p_H$: 0.154 | $p$: 0.745; $p_H$: 1.000 |
| Not observed in 2012 | 42.752 [19.351, 66.153] | 0.779 [-1.036, 2.594] | 39.928 [7.740, 72.117] | -0.053 [-0.414, 0.309] |
| $N$=62; states=12; eff.$N$=35.8 | $p$: 0.027; $p_H$: 0.137 | $p$: 0.380; $p_H$: 0.893 | $p$: 0.174; $p_H$: 0.692 | $p$: 0.774; $p_H$: 1.000 |
| | -25.138 [-54.965, 4.690] | 1.444 [-0.170, 3.059] | -33.056 [-66.049, -0.063] | 0.054 [-0.303, 0.411] |
| Old minus recent | $p$: 0.195; $p_H$: 0.901 | $p$: 0.079; $p_H$: 0.707 | $p$: 0.122; $p_H$: 1.000 | $p$: 0.769; $p_H$: 1.000 |
| | 11.265 [-23.592, 46.123] | 1.613 [-0.027, 3.254] | 16.983 [-17.941, 51.906] | -0.110 [-0.488, 0.268] |
| Recent minus 2009–10 | $p$: 0.643; $p_H$: 1.000 | $p$: 0.100; $p_H$: 0.796 | $p$: 0.478; $p_H$: 1.000 | $p$: 0.606; $p_H$: 1.000 |
| | -15.843 [-32.837, 1.151] | -0.116 [-2.372, 2.139] | -50.432 [-100.579, -0.285] | -0.375 [-0.706, -0.043] |
| 2009–10 minus 2011–12 | $p$: 0.152; $p_H$: 0.901 | $p$: 0.930; $p_H$: 1.000 | $p$: 0.276; $p_H$: 1.000 | $p$: 0.166; $p_H$: 1.000 |
| | 35.837 [-0.279, 71.953] | -1.275 [-2.732, 0.182] | 56.697 [-17.497, 130.892] | 0.110 [-0.316, 0.536] |
| 2011–12 minus not observed | $p$: 0.193; $p_H$: 0.901 | $p$: 0.110; $p_H$: 0.796 | $p$: 0.353; $p_H$: 1.000 | $p$: 0.650; $p_H$: 1.000 |
| | 29.715 [2.313, 57.118] | -2.941 [-4.325, -1.558] | 66.505 [12.767, 120.243] | 0.431 [-0.049, 0.910] |
| 2011–12 minus baseline old | $p$: 0.150; $p_H$: 0.901 | $p$: 0.023; $p_H$: 0.228 | $p$: 0.161; $p_H$: 1.000 | $p$: 0.164; $p_H$: 1.000 |

*Notes:* Cells give quota-response estimates and state-clustered 95% intervals, followed by 9,999-draw wild $p$ and Holm-adjusted $p_H$. Models use the fixed headline quota standard deviation, municipality/year effects, full baseline controls and population weights unless noted. Group models fully interact all nuisance terms. Receipt/votes and registration/turnout define separate primary/secondary families; slopes and contrasts are adjusted separately. Groups with fewer than ten states or 25 municipalities retain estimates but no inferential interval or bootstrap test. Registrations are retrospective 2009–2012 first registrations observed in the 2012 snapshot. Later cohorts use complete opening years of units observed in 2012 where canonical 2008 CRAS status was zero. Unknown dates or contradictory pre-2009 dates are not forced into cohorts. Not observed is census file membership, not verified absence. The late cohort ends in September 2012; October–December and unknown 2012 opening months are excluded and audited. Opening dates contradicted by an earlier operating observation are excluded. Cohorts still condition on post-treatment institutional choices. Adjacent comparisons plus late-versus-old form a ten-test primary family and a ten-test secondary family. These comparisons do not establish causal duration or monotonicity.

**Figure 4:** Electoral Responses and CRAS Institutional Exposure




*Notes:* Points show incumbent vote-share responses, in percentage points, to one fixed headline-sample population-weighted standard deviation of the cumulative quota shock. Bars are pointwise state-clustered confidence intervals. Panel A reports adoption-cohort estimates; Panel B reports older-minus-recent baseline contrasts under population and overlap weights. The post-2008 opening cohorts condition on offices reported in the 2012 census; later entry is a post-reform characteristic. The last group is not observed in that census, which does not establish the absence of an office. The late-opening cohort ends before the 2012 election. Tables 24 and 25 report support, contrasts and adjusted tests. These comparisons do not identify a causal effect of operating duration.

**Table 26:** Census Retention of Municipalities With a Baseline CRAS

| Endpoint | Not observed (1) | States (2) | Initial quota RF (3) | Holm p (4) | Cumulative quota RF (5) | Holm p (6) |
|---|---|---|---|---|---|---|
| 2009 | 14 | 6 | 0.112 (0.079)<br>[-0.051, 0.275] | 1.000 | 0.180 (0.108)<br>[-0.042, 0.403] | 1.000 |
| 2010 | 8 | 5 | 0.050 (0.178)<br>[-0.316, 0.417] | 1.000 | 0.021 (0.112)<br>[-0.211, 0.252] | 1.000 |
| 2011 | 24 | 12 | -0.040 (0.249)<br>[-0.552, 0.473] | 1.000 | -0.332 (0.154)<br>[-0.649, -0.016] | 0.280 |
| 2012 | 28 | 9 | 0.067 (0.492)<br>[-0.946, 1.079] | 1.000 | -0.143 (0.347)<br>[-0.857, 0.571] | 1.000 |

*Notes:* All 1394 municipalities have a CRAS in the canonical 2008 source. Not observed means no office row in the endpoint census; it does not establish closure. States counts states containing at least one such municipality. The outcome is census retention in percentage points, with baseline retention fixed at 100. A positive coefficient indicates greater retention. Models retain municipality/year effects, the full baseline control specification, party-by-post terms and 2006 population weights; standard errors are clustered across 26 states. Parentheses give standard errors; brackets give 95% CRV1 t intervals. Both instruments use their own full-sample weighted SD, held fixed across endpoints: 158.883 and 171.496 quota slots per 10,000 residents. Eight null-imposed wild-cluster tests (9,999 Rademacher draws) share a Holm family. Sparse census losses may provide little effective cluster support even though the regression contains 26 states. This diagnostic concerns selection into observed-count regressions. The cumulative instrument is a consistency comparison, not evidence that future quota decisions caused earlier census participation.

**Table 27:** Quota Exposure and the Observed CRAS Network

| | Initial 2009 quota | Cumulative 2012 quota | Municipalities / states |
|---|---|---|---|
| *Change in observed unit count* | | | |
| 2009 | -0.160 [-0.292, -0.028]<br>$p$: 0.085; $p_H$: 1.000 | -0.059 [-0.189, 0.071]<br>$p$: 0.508; $p_H$: 1.000 | 1,380 / 26 |
| 2010 | -0.217 [-0.393, -0.041]<br>$p$: 0.038; $p_H$: 0.720 | -0.257 [-0.413, -0.101]<br>$p$: 0.002; $p_H$: 0.046 | 1,386 / 26 |
| 2011 | -0.197 [-0.394, 0.001]<br>$p$: 0.053; $p_H$: 0.950 | -0.113 [-0.255, 0.029]<br>$p$: 0.095; $p_H$: 1.000 | 1,370 / 26 |
| 2012 | -0.287 [-0.511, -0.063]<br>$p$: 0.035; $p_H$: 0.700 | -0.218 [-0.397, -0.039]<br>$p$: 0.021; $p_H$: 0.443 | 1,366 / 26 |
| *Reported post-2008 openings* | | | |
| 2009 | -0.124 [-0.230, -0.018]<br>$p$: 0.109; $p_H$: 1.000 | -0.032 [-0.143, 0.078]<br>$p$: 0.691; $p_H$: 1.000 | 1,595 / 26 |
| 2010 | -0.147 [-0.223, -0.071]<br>$p$: 0.018; $p_H$: 0.400 | -0.225 [-0.331, -0.118]<br>$p$: 0.003; $p_H$: 0.076 | 1,758 / 26 |
| 2011 | -0.066 [-0.185, 0.053]<br>$p$: 0.321; $p_H$: 1.000 | -0.033 [-0.155, 0.089]<br>$p$: 0.650; $p_H$: 1.000 | 1,747 / 26 |
| 2012 | -0.148 [-0.326, 0.031]<br>$p$: 0.220; $p_H$: 1.000 | -0.139 [-0.316, 0.039]<br>$p$: 0.244; $p_H$: 1.000 | 1,826 / 26 |
| *Endpoint census presence among baseline-coded no CRAS (pp)* | | | |
| 2009 | 0.923 [-3.361, 5.206]<br>$p$: 0.685; $p_H$: 1.000 | 0.106 [-2.693, 2.906]<br>$p$: 0.940; $p_H$: 1.000 | 684 / 22 |
| 2010 | -0.828 [-4.769, 3.112]<br>$p$: 0.675; $p_H$: 1.000 | 0.577 [-2.937, 4.090]<br>$p$: 0.748; $p_H$: 1.000 | 684 / 22 |
| 2011 | -0.983 [-3.040, 1.073]<br>$p$: 0.452; $p_H$: 1.000 | -0.796 [-2.386, 0.793]<br>$p$: 0.338; $p_H$: 1.000 | 684 / 22 |
| 2012 | 0.775 [-0.564, 2.114]<br>$p$: 0.287; $p_H$: 1.000 | -0.564 [-1.753, 0.624]<br>$p$: 0.326; $p_H$: 1.000 | 684 / 22 |

*Notes:* Each cell gives a reduced-form coefficient, a state-clustered 95% interval in brackets, the 9,999-draw wild-cluster $p$-value and its Holm adjustment across all 24 declared tests. Each instrument is divided by its own fixed full-sample population-weighted standard deviation. Initial 2009 exposure precedes the 2010–2012 outcomes, without resolving within-2009 timing; cumulative 2012 exposure is not a pre-2009 assignment measure. Count changes require observed unit records in both 2008 and the endpoint year. Opening counts refer to units still observed at that endpoint with valid reported opening years after 2008 and no earlier observation as an operating unit contradicting that date; they are not all historical openings. The final panel is the probability of appearing in the endpoint census among municipalities with canonical baseline CRAS indicator zero; its zeros mean not observed, not verified absence. Models use the full baseline specification, 2006-population weights and municipality/year fixed effects on the reported subset of the locked panel. Missing responses are not set to zero counts. Imprecise intervals do not establish infrastructure exclusion.

**Table 28:** Where the Electoral Response Appears: Additive Reduced Forms

| | Pooled | No CRAS | CRAS | Difference |
|---|---|---|---|---|
| Incumbent votes | 1.145 | -0.204 | 1.486 | 1.690 |
| State-clustered s.e. | (0.407) | (0.488) | (0.443) | (0.505) |
| Wild-cluster $p$ | 0.052 | 0.699 | 0.019 | 0.021 |
| Holm-adjusted $p$ | 0.104 | 1.000 | 0.056 | 0.103 |
| Other candidates' votes | -1.205 | 0.331 | -1.566 | -1.897 |
| State-clustered s.e. | (0.391) | (0.530) | (0.452) | (0.671) |
| Wild-cluster $p$ | 0.022 | 0.635 | 0.008 | 0.045 |
| Holm-adjusted $p$ | 0.073 | 1.000 | 0.033 | 0.182 |
| Blank or null ballots | -0.294 | -0.265 | -0.305 | -0.040 |
| State-clustered s.e. | (0.194) | (0.189) | (0.252) | (0.302) |
| Wild-cluster $p$ | 0.100 | 0.177 | 0.232 | 0.898 |
| Holm-adjusted $p$ | 0.104 | 0.530 | 0.232 | 1.000 |
| Abstention | 0.354 | 0.138 | 0.385 | 0.247 |
| State-clustered s.e. | (0.180) | (0.073) | (0.213) | (0.206) |
| Wild-cluster $p$ | 0.018 | 0.085 | 0.040 | 0.309 |
| Holm-adjusted $p$ | 0.073 | 0.341 | 0.079 | 0.926 |
| Registered voters per 10,000 residents | 120.393 | 110.898 | 124.979 | 14.080 |
| State-clustered s.e. | (34.967) | (22.257) | (40.831) | (38.417) |
| Wild-cluster $p$ | $< 0.001$ | 0.002 | $< 0.001$ | 0.735 |
| Holm-adjusted $p$ | 0.001 | 0.010 | 0.004 | 1.000 |
| System municipalities | 2,078 | 2,078 | 2,078 | 2,078 |
| State clusters | 26 | 26 | 26 | 26 |

*Notes:* Each coefficient is the response to one fixed, headline-sample weighted standard deviation of the quota shock. The first four outcomes are percentages of registered voters in the same municipal election and sum to 100; their coefficients sum to zero within each column. Other candidates' votes include every valid vote not cast for the 2008 winner. Blank or null ballots are ballots cast minus valid candidate votes. The final outcome uses fixed 2006 population instead of the electorate. All models retain the locked 2008–2012 panel, municipality and year fixed effects, full baseline controls and 2006-population weights. CRAS columns fully interact the controls, party-by-post effects and post indicator with 2008 CRAS status; Difference is CRAS minus no CRAS from that pooled system. The pooled column imposes common control slopes. Wild-cluster tests use 9,999 null-imposed Rademacher draws clustered by state. Holm adjustments cover all five outcomes separately within each column. The Difference column tests the CRAS contrasts; group columns are not independent tests. Aggregate accounting cannot identify individual voter transitions or prove political persuasion.

**Table 29:** Selection into Rerunning

| | Incumbent Runs Again in 2012 | |
|---|---|---|
| | (1) | (2) |
| $\Delta$Quota$_m$ | -2.183 | 0.045 |
| | ( 1.156 ) | ( 1.017 ) |
| Incumbent Party | ✓ | ✓ |
| *Bolsa Família* Intensity | ✓ | ✓ |
| Municipality Characteristics | | ✓ |
| Mayor Characteristics | | ✓ |
| Number of Mayors / Municipalities | 5,492 | 5,492 |
| Observations | 5,492 | 5,492 |

*Notes:* The dependent variable equals 100 if the mayor elected in 2008 appears on the 2012 ballot and zero otherwise. The estimation universe is constructed before conditioning on rerunning. Every column is weighted by 2006 population and includes state fixed effects because municipality and year effects are not identified in this one-observation-per-mayor design. The quota shock is a rate per 10,000 residents using fixed 2006 population and is divided by the main-sample analysis-weighted SD. Column (1) includes incumbent-party effects and baseline *Bolsa Família* intensity. Column (2) adds municipality and incumbent-mayor characteristics. These controls are baseline and pre-shock: party, program intensity, and mayor characteristics use 2008 values; population uses 2006; and the remaining municipality characteristics use the 2000 Census. Both columns therefore use only pre-shock controls. Standard errors clustered by state are reported in parentheses. A † attached to a standard error denotes a wild-cluster $p < 0.05$, with resampling by state.

**Table 30:** Pre-Reform Vote-Share Balance

| | Incumbent's Vote Share in 2008 | |
|---|---|---|
| | (1) | (2) |
| $\Delta$Quota$_m$ | 0.127 | -0.070 |
| | ( 0.254 ) | ( 0.228 ) |
| Incumbent Party | ✓ | ✓ |
| *Bolsa Família* Intensity | ✓ | ✓ |
| Municipality Characteristics | | ✓ |
| Mayor Characteristics | | ✓ |
| Number of Mayors / Municipalities | 2,078 | 2,078 |
| Observations | 2,078 | 2,078 |

*Notes:* The dependent variable is the vote share of the mayor elected in 2008, measured before the quota reform. Every column is weighted by 2006 population and includes state fixed effects. The quota shock is a rate per 10,000 residents using fixed 2006 population and is divided by its main-sample analysis-weighted SD. Column (1) includes incumbent-party effects and baseline *Bolsa Família* intensity. Column (2) adds municipality and incumbent-mayor characteristics. These controls are baseline and pre-shock: party, program intensity, and mayor characteristics use 2008 values; population uses 2006; and the remaining municipality characteristics use the 2000 Census. Both columns therefore use only pre-shock controls. Standard errors clustered by state are reported in parentheses. A † attached to a standard error denotes a wild-cluster $p < 0.05$, with resampling by state.

**Table 31:** Alternative Exclusion Channels

| | Alternative Exclusion Channels | |
|---|---|---|
| | (1) | (2) |
| *Panel A: Δ Fundo de Participação dos Municípios (FPM) Transfers* | | |
| $\Delta\text{Quota}_m$ | 0.082 | 0.107 |
| | ( 0.058 ) | ( 0.020 )† |
| *Panel B: Δ Per-Family Bolsa Família Benefit Value* | | |
| $\Delta\text{Quota}_m$ | 0.049 | 0.083 |
| | ( 0.043 ) | ( 0.035 ) |
| Incumbent Party | ✓ | ✓ |
| *Bolsa Família* Intensity | ✓ | ✓ |
| Municipality Characteristics | | ✓ |
| Mayor Characteristics | | ✓ |
| Number of Mayors / Municipalities | 2,078 | 2,078 |
| Observations | 4,156 | 4,156 |

*Notes:* Panel A uses the change between the 12-month average of total municipal FPM transfers in 2008 and 2012, per 10,000 residents using fixed 2006 population, divided by its full-sample 2006-population-weighted SD. Panel B uses the change between the simple averages of the twelve monthly per-family benefit values in 2008 and 2012, divided by the cross-municipality SD of its change. The analysis-weighted mean changes are R$57,959.64 per month per 10,000 residents for FPM and R$49.48 per family per month for benefit value. Multiplying the standardized Column (2) quota coefficients by the corresponding outcome SD gives R$5,466.59 per month per 10,000 residents and R$0.91 per family per month, respectively. All models use 2006-population weights. Each displayed row is a reduced-form channel diagnostic and is not conditioned on realized beneficiary expansion. Column (1) includes incumbent-party effects and baseline *Bolsa Família* intensity. Column (2) adds municipality and incumbent-mayor characteristics. These controls are baseline and pre-shock: party, program intensity, and mayor characteristics use 2008 values; population uses 2006; and the remaining municipality characteristics use the 2000 Census. Both columns therefore use only pre-shock controls. Standard errors clustered by state are reported in parentheses. A † attached to a standard error denotes a wild-cluster $p < 0.05$, with resampling by state.

**Table 32:** Reverse Causality Channels

| | Reverse Causality Channels | | |
|---|---|---|---|
| | (1) | (2) | (3) |
| *Panel A: $\Delta$ Permitted Bolsa Família Expansion* | | | |
| Incumbent's Vote Share in 2008 | 0.009 | -0.005 | -0.008 |
| | ( 0.018 ) | ( 0.016 ) | ( 0.019 ) |
| *Panel B: $\Delta$ Realized Bolsa Família Expansion* | | | |
| Incumbent's Vote Share in 2008 | -0.058 | -0.049 | -0.042 |
| | ( 0.026 )† | ( 0.022 ) | ( 0.019 )† |
| Incumbent Party | ✓ | ✓ | ✓ |
| *Bolsa Família* Intensity | ✓ | ✓ | ✓ |
| Municipality Characteristics | | ✓ | ✓ |
| Mayor Characteristics | | ✓ | ✓ |
| Benefit Value | | | ✓ |
| *FPM* Transfers | | | ✓ |
| Number of Mayors / Municipalities | 2,078 | 2,078 | 2,078 |
| Observations | 2,078 | 2,078 | 2,078 |

*Notes:* Panel A measures permitted quota expansion in quota-shock-SD units. Panel B measures the 2008–2012 net change in the 12-month average beneficiary-family stock in treatment-change-SD units. The regressor is the 2008 winner's vote share divided by its analysis-weighted SD of 9.59 percentage points. Every column is weighted by 2006 population and includes state fixed effects. Columns (1)–(2) follow the pre-treatment control sequence. Column (3) additionally includes 2008 benefit value and FPM transfers. Standard errors clustered by state are reported in parentheses. A † attached to a standard error denotes a wild-cluster $p < 0.05$, with resampling by state.

**Table 33:** Monotonicity by Quota-Shock Size

| | $\Delta$Beneficiaries$_m$ | | |
|---|---|---|---|
| | (1) | (2) | (3) |
| *Panel A: Top 50%* | | | |
| $\Delta$Quota$_m$ | 0.504 | 0.545 | 0.527 |
| | ( 0.062 )† | ( 0.057 )† | ( 0.056 )† |
| *Panel B: Top 25%* | | | |
| $\Delta$Quota$_m$ | 0.485 | 0.518 | 0.531 |
| | ( 0.091 )† | ( 0.071 )† | ( 0.066 )† |
| *Panel C: Top 15%* | | | |
| $\Delta$Quota$_m$ | 0.331 | 0.440 | 0.494 |
| | ( 0.153 ) | ( 0.116 )† | ( 0.099 )† |
| *Panel D: Top 5%* | | | |
| $\Delta$Quota$_m$ | 0.510 | 0.604 | 0.630 |
| | ( 0.181 ) | ( 0.124 )† | ( 0.104 )† |
| *Panel E: Top 1%* | | | |
| $\Delta$Quota$_m$ | 0.776 | 0.801 | 0.828 |
| | ( 0.163 ) | ( 0.135 ) | ( 0.128 )† |
| Incumbent Party | ✓ | ✓ | ✓ |
| *Bolsa Família* Intensity | ✓ | ✓ | ✓ |
| Municipality Characteristics | | ✓ | ✓ |
| Mayor Characteristics | | ✓ | ✓ |
| Benefit Value | | | ✓ |
| *FPM* Transfers | | | ✓ |

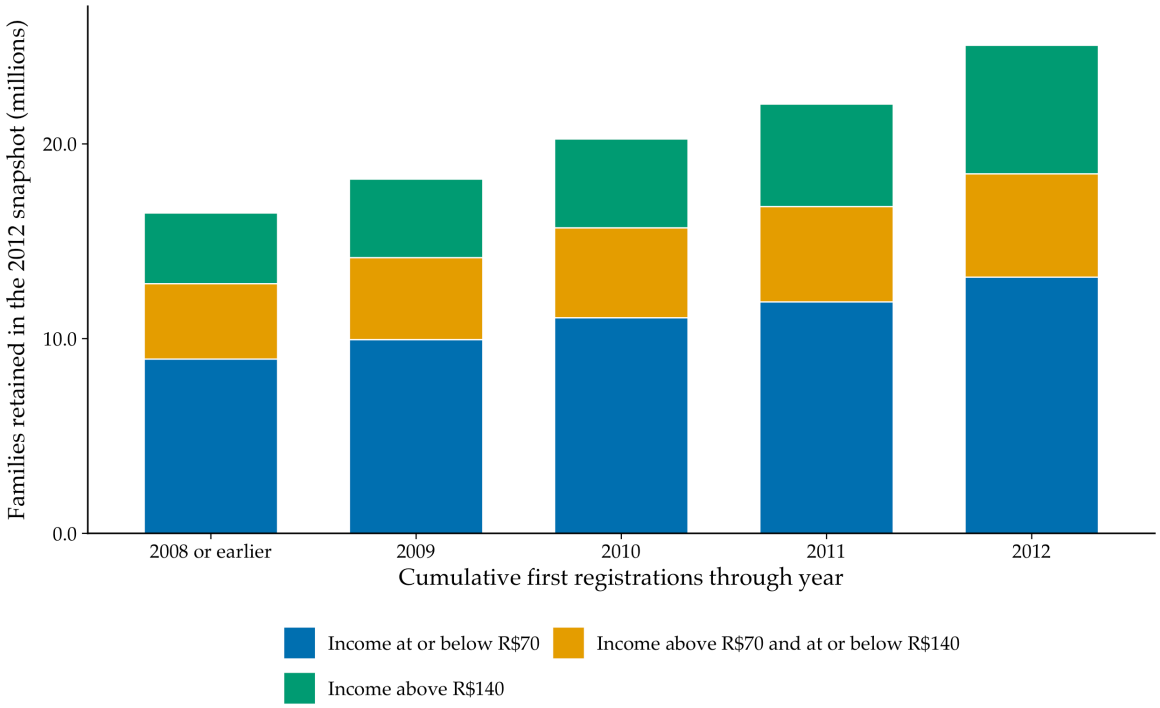
*Notes:* Each panel estimates separate slopes for the named upper-tail group and its complement on the full locked sample; the table reports the upper-tail slope. Groups are defined by unweighted quota-shock quantiles and are nested. Columns follow the headline control sequence, including baseline benefit value and FPM transfers in Column (3), with 2006-population weights and state-clustered standard errors. † denotes the wild-cluster result. This is an average first-stage-shape diagnostic, not a test of unit-level monotonicity.

**Table 34:** Family Income in 2012 by First-Registration Cohort

| | Registered before 2009 | | Registered in 2009–2012 | |
|---|---|---|---|---|
| | Mean | SD | Mean | SD |
| *Panel A: Monthly per-capita family income indicators* | | | | |
| Income at or below R$70 | 0.544 | 0.498 | 0.489 | 0.500 |
| Income above R$70 through R$140 | 0.235 | 0.424 | 0.167 | 0.373 |
| Income above R$140 | 0.221 | 0.415 | 0.345 | 0.475 |
| Missing income | 0.000 | 0.000 | 0.000 | 0.000 |
| *Panel B: Cohort size* | | | | |
| Families | 16,452,008 | | 8,612,498 | |

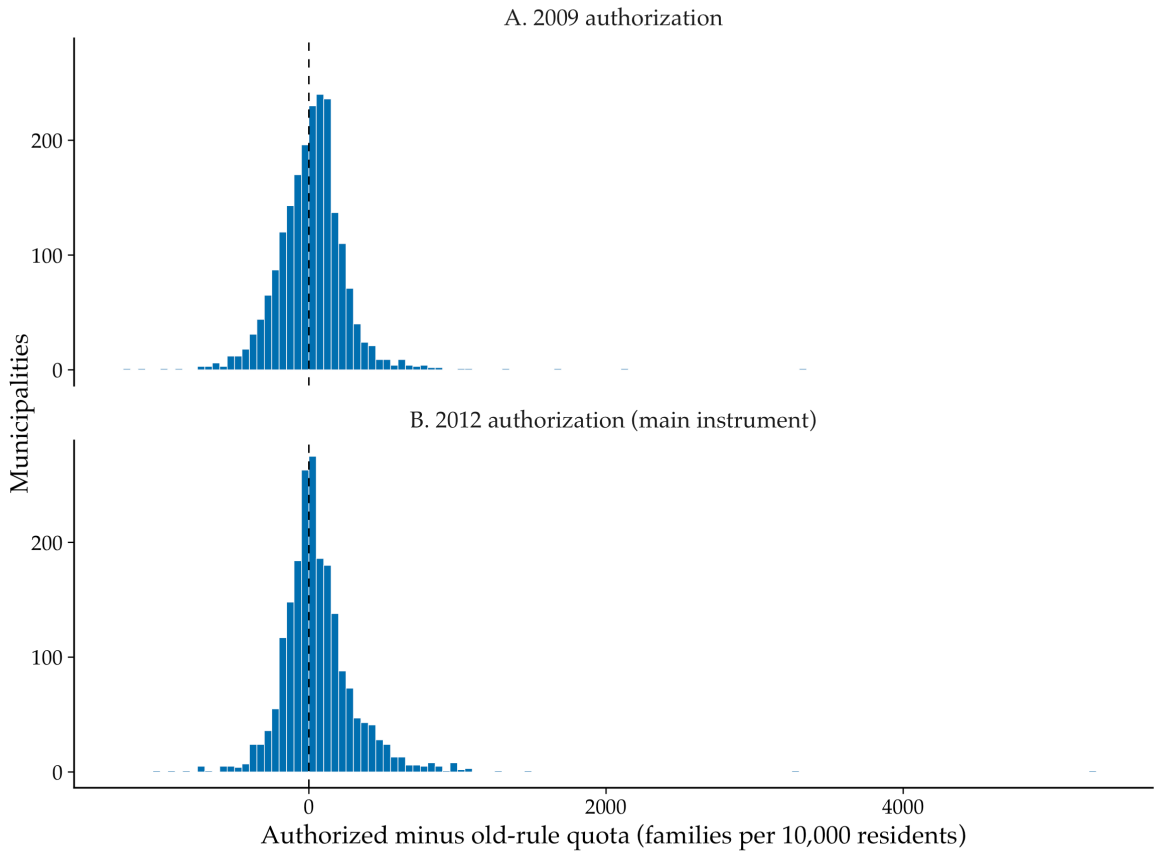
*Notes:* National, unweighted family-level composition among registrations retained in the 2012 *Cadastro Único* snapshot with valid first-registration dates. All income measures refer to 2012; the first column group is not a 2009 income snapshot. Means are shares, and SD is the sample standard deviation of the corresponding binary indicator. Income groups are mutually exclusive; zero reported income is included in the lowest band. Cohorts include all registered families and do not identify actual *Bolsa Família* receipt. The source includes two unresolved municipality tokens retained in national totals. Invalid registration dates are excluded as documented in the source audit. Individual characteristics, mean income, and household composition cannot be recovered from these grouped counts.

**Figure 5:** Cumulative Family Registrations Observed in the 2012 *Cadastro Único* Snapshot, by Reported Income



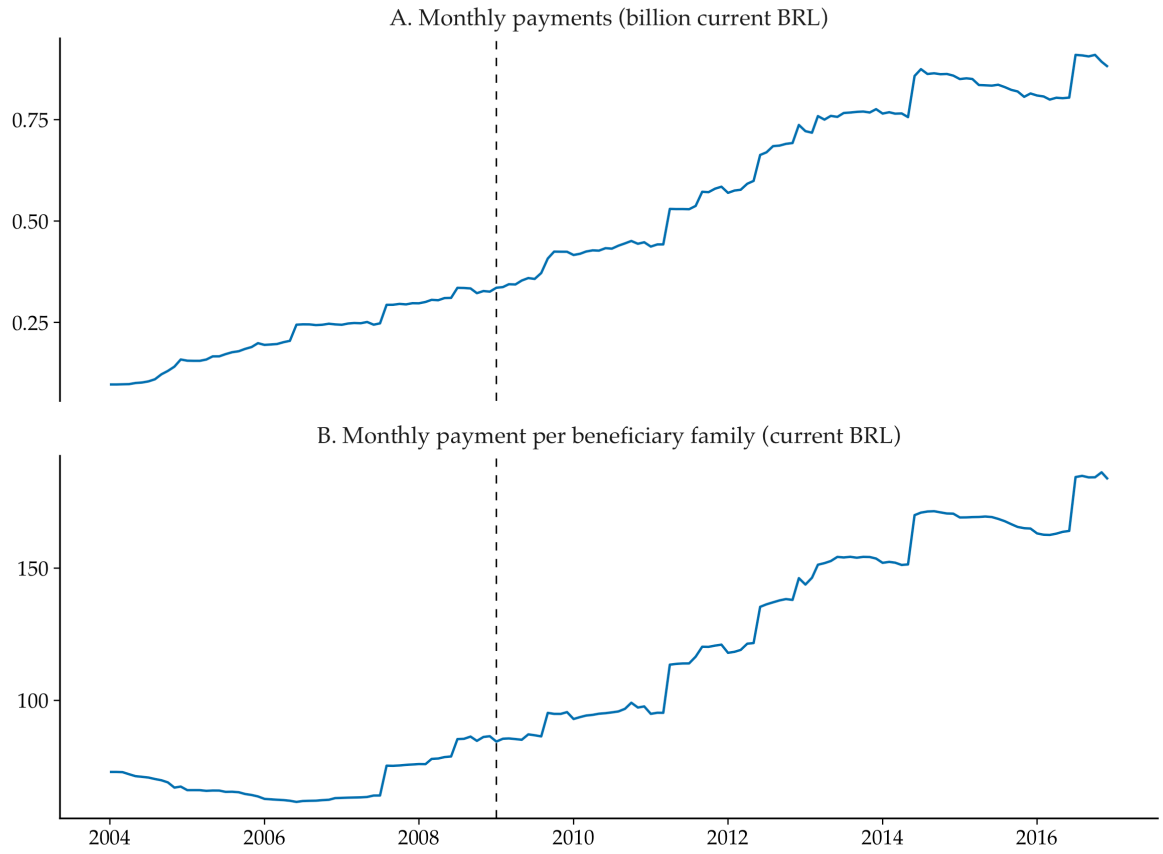

*Notes:* Counts sum municipality-level family records in the 2012 snapshot. "2008 or earlier" pools all valid registration dates through December 2008. Municipality and nominal monthly per-capita family income are recorded in 2012; income is not measured at registration. Registration dates outside 1900–2012 are excluded. Each bar cumulates families through the indicated registration year, conditional on their presence in the 2012 snapshot. Stacked segments are mutually exclusive 2012 income categories: at most R$70, above R$70 through R$140, above R$140, and missing or invalid income. These are retrospective cohorts, not historical registry stocks. The available aggregates pool years through 2008 and contain no registrations after 2012; they cannot support separate 2004–2008 bars or a series through 2016.

**Figure 6:** Municipal Exposure to the Quota Reallocation



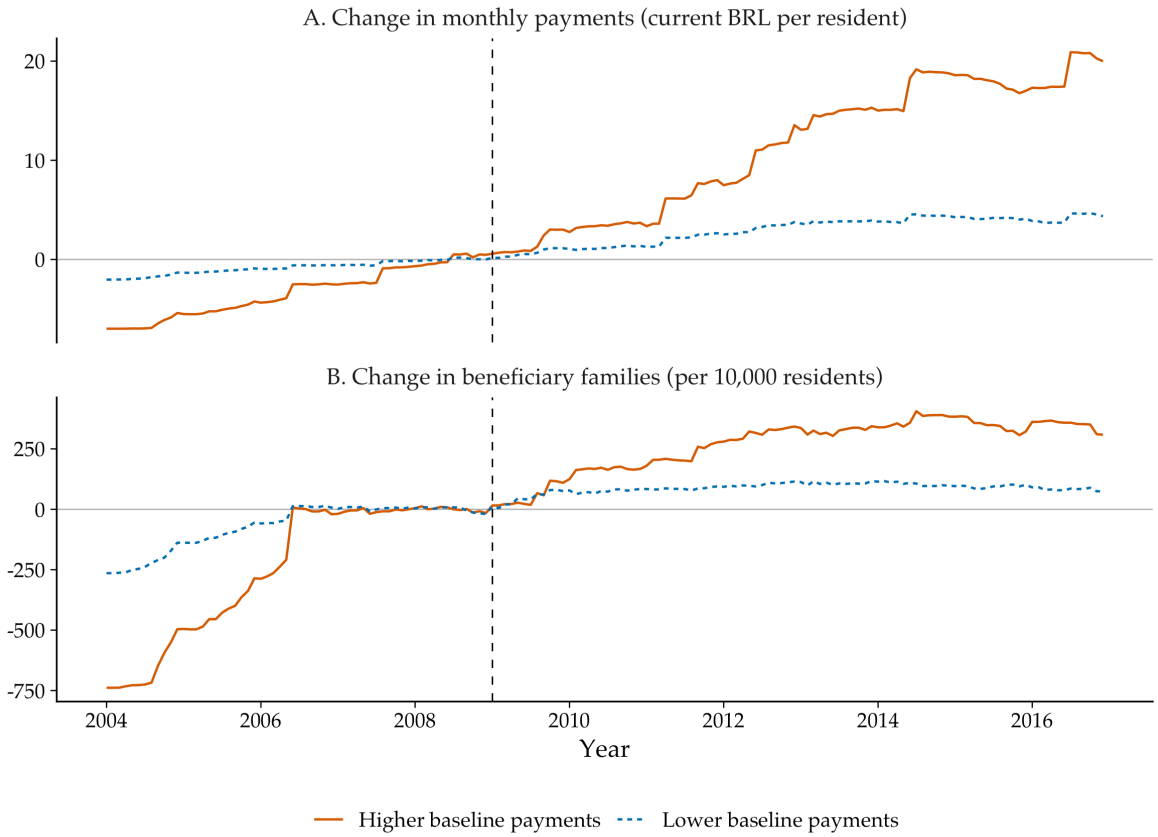

*Notes:* Unweighted distributions for the unchanged headline sample of 2,078 municipalities. Exposure is official 2009 or 2012 authorization minus the counterfactual 2009 quota, per 10,000 residents using fixed 2006 population. The counterfactual uses the previous allocation formula with the national expansion and R$90/137 poverty cutoffs; it is not a no-program counterfactual. All tails are retained. The 2012 panel shows the main instrument before standard-deviation scaling. Source: MDS quota records and the project's counterfactual construction.

**Figure 7:** Monthly *Bolsa Família* Payments, 2004–2016




*Notes:* Totals cover the same 2,078 headline municipalities throughout, rather than Brazil as a whole. Payments are nominal BRL; the mean benefit is aggregate payments divided by beneficiary families. The series retains the existing data-construction corrections for isolated payment spikes. The dashed line marks the year 2009, not the implementation month. Source: MDS monthly municipal program records, distinct from the 2012 family-registration snapshot.

**Figure 8:** Program Expansion by Baseline Payment Intensity




*Notes:* Municipalities are divided at the unweighted median of 2008 monthly payments per 2006 resident (BRL 6.55). Within each group, payments and beneficiary-family counts are divided by the group's fixed 2006 population. Each series subtracts its own 2008 calendar-year mean. Monetary values are nominal. These profiles describe heterogeneity in program expansion; they do not estimate the causal effect of baseline payment intensity or an electoral return per real. Source: MDS municipal program records.

**Table 35:** Growth in *Bolsa Família* Payments: An Accounting Decomposition

| | All municipalities | Lower baseline payments | Higher baseline payments |
|---|---|---|---|
| Municipalities | 2,078 | 1,039 | 1,039 |
| Payments, 2008 (million BRL) | 317.29 | 126.96 | 190.33 |
| Payments, 2012 (million BRL) | 644.65 | 271.80 | 372.85 |
| Payment change (million BRL) | 327.36 | 144.85 | 182.52 |
| Family-count component | 113.06 | 48.95 | 64.12 |
| Mean-benefit component | 214.30 | 95.90 | 118.40 |
| Mean benefit, 2008 (BRL) | 81.60 | 74.41 | 87.22 |
| Mean benefit, 2012 (BRL) | 130.76 | 124.47 | 135.76 |

Notes: The unchanged headline sample contains 2,078 municipalities. Groups divide municipalities at the unweighted median of 2008 monthly payments per 2006 resident (BRL 6.55). Payments and family counts are calendar-year means of the monthly MDS administrative series. All values are in current BRL. Both payment-change components are in million BRL; mean benefits are per beneficiary family. For municipality $m$, write $V_{mt} = N_{mt}b_{mt}$, where $N$ is beneficiary families and $b$ is payments per family. The quantity component is $(b_{m,2012} + b_{m,2008})(N_{m,2012} - N_{m,2008})/2$; the benefit component is $(N_{m,2012} + N_{m,2008})(b_{m,2012} - b_{m,2008})/2$. These components sum exactly to the payment change before rounding. They describe accounting changes, not causal effects. The benefit component combines nominal benefit adjustments and changes in beneficiary composition. Source: MDS municipal program records.

**Table 36:** Quota Expansion and Family Registration

| | New registrations per 10,000 residents | | |
|---|---|---|---|
| | (1) | (2) | (3) |
| All registered families | 29.200 | 38.371 | 40.187 |
| | ( 9.554 ) | ( 9.955 ) | ( 9.786 ) |
| Kleibergen–Paap $F$ | 9.34 | 14.86 | 16.87 |
| Income above R$140 | -4.131 | -1.589 | 4.163 |
| | ( 4.545 ) | ( 3.976 ) | ( 3.625 ) |
| Kleibergen–Paap $F$ | 0.83 | 0.16 | 1.32 |
| Income at or below R$140 | 33.331 | 39.960 | 36.024 |
| | ( 8.558 ) | ( 8.800 ) | ( 8.146 ) |
| Kleibergen–Paap $F$ | 15.17 | 20.62 | 19.55 |
| Income at or below R$70 | 30.449 | 35.883 | 30.017 |
| | ( 7.547 ) | ( 7.592 ) | ( 6.624 ) |
| Kleibergen–Paap $F$ | 16.28 | 22.34 | 20.53 |
| Income above R$70 and at or below R$140 | 2.882 | 4.077 | 6.007 |
| | ( 1.392 ) | ( 1.774 ) | ( 2.020 ) |
| Kleibergen–Paap $F$ | 4.29 | 5.28 | 8.84 |
| Incumbent party and baseline *Bolsa Família* intensity | ✓ | ✓ | ✓ |
| Municipality and mayor characteristics | | ✓ | ✓ |
| Benefit value and *FPM* transfers | | | ✓ |
| Number of Municipalities | 2,078 | 2,078 | 2,078 |
| Observations | 4,156 | 4,156 | 4,156 |

*Notes:* All columns retain the locked 2008–2012 sample, quota-shock instrument, municipality and year fixed effects, 2006-population weights and state-clustered CRV1 standard errors. Controls are interacted with post. Registration expansion counts families first registered in 2009–2012 and still observed in the 2012 *Cadastro Único* snapshot. Income (monthly per capita) and municipality are measured in 2012. These are surviving entry cohorts, not historical net stocks or verified *Bolsa Família* recipients. The income definitions overlap: the at-or-below-R$140 group combines the two lower-income bands. Each coefficient reports the change in family registrations per 10,000 residents (fixed 2006 population) induced by one weighted SD of the quota shock. Natural-rate first stages add across disjoint income groups. Parentheses report standard errors.

**Table 37:** Quota Expansion and Electoral Outcomes

| | Electoral outcomes (percentage points) | | |
|---|---|---|---|
| | (1) | (2) | (3) |
| Incumbent vote share | 1.929 | 1.299 | 1.414 |
| | ( 0.604 )† | ( 0.461 )† | ( 0.482 )† |
| Turnout | -0.372 | -0.352 | -0.354 |
| | ( 0.192 )† | ( 0.181 )† | ( 0.180 )† |
| Incumbent party and baseline *Bolsa Família* intensity | ✓ | ✓ | ✓ |
| Municipality and mayor characteristics | | ✓ | ✓ |
| Benefit value and *FPM* transfers | | | ✓ |
| Number of Municipalities | 2,078 | 2,078 | 2,078 |
| Observations | 4,156 | 4,156 | 4,156 |

*Notes:* All columns retain the locked 2008–2012 sample, quota-shock instrument, municipality and year fixed effects, 2006-population weights and state-clustered CRV1 standard errors. Controls are interacted with post. Registration expansion counts families first registered in 2009–2012 and still observed in the 2012 *Cadastro Único* snapshot. Income (monthly per capita) and municipality are measured in 2012. These are surviving entry cohorts, not historical net stocks or verified *Bolsa Família* recipients. The income definitions overlap: the at-or-below-R$140 group combines the two lower-income bands. These reduced forms are identical for every registration definition and reproduce the headline reduced forms; they are reported once. Parentheses report standard errors. † denotes a state wild-cluster p-value below 5% using 9,999 Rademacher draws.

**Table 38:** Registration Expansion and Incumbent Vote Share

| | Incumbent vote share (percentage points) | | |
|---|---|---|---|
| | (1) | (2) | (3) |
| All registered families | 15.773 | 8.085 | 8.401 |
| | ( 7.453 )† | ( 3.933 )† | ( 4.039 )† |
| AR 95% confidence set | [4.819, 55.175] | [1.867, 22.852] | [2.076, 23.020] |
| Income above R$140 | -50.197 | -87.908 | 36.512 |
| | ( 56.262 )† | ( 225.199 )† | ( 34.739 )† |
| AR 95% confidence set | $(-\infty$, -10.938] ∪ [32.501, $\infty)$ | $(-\infty$, -8.061] ∪ [17.491, $\infty)$ | $(-\infty$, -42.334] ∪ [7.205, $\infty)$ |
| Income at or below R$140 | 11.680 | 6.562 | 7.922 |
| | ( 4.864 )† | ( 3.099 )† | ( 3.765 )† |
| AR 95% confidence set | [3.772, 29.293] | [1.512, 16.719] | [1.954, 20.788] |
| Income at or below R$70 | 12.368 | 7.069 | 9.197 |
| | ( 5.082 )† | ( 3.265 )† | ( 4.224 )† |
| AR 95% confidence set | [4.011, 30.094] | [1.655, 17.378] | [2.338, 23.106] |
| Income above R$70 and at or below R$140 | 28.690 | 13.661 | 10.090 |
| | ( 16.772 )† | ( 8.816 )† | ( 5.970 )† |
| AR 95% confidence set | [8.204, 5774.366] | [2.664, 165.827] | [2.129, 46.582] |
| Incumbent party and baseline *Bolsa Família* intensity | ✓ | ✓ | ✓ |
| Municipality and mayor characteristics | | ✓ | ✓ |
| Benefit value and *FPM* transfers | | | ✓ |
| Number of Municipalities | 2,078 | 2,078 | 2,078 |
| Observations | 4,156 | 4,156 | 4,156 |

*Notes:* All columns retain the locked 2008–2012 sample, quota-shock instrument, municipality and year fixed effects, 2006-population weights and state-clustered CRV1 standard errors. Controls are interacted with post. Registration expansion counts families first registered in 2009–2012 and still observed in the 2012 *Cadastro Único* snapshot. Income (monthly per capita) and municipality are measured in 2012. These are surviving entry cohorts, not historical net stocks or verified *Bolsa Família* recipients. The income definitions overlap: the at-or-below-R$140 group combines the two lower-income bands. Each row uses a separate endogenous registration measure, scaled by its own 2006-population-weighted SD of the change in registrations per 10,000 residents. Coefficients are not directly comparable across row-specific SDs. One common instrument does not identify separate causal effects by family income or the effect of registration apart from subsequent transfers. First-stage diagnostics appear in Table 36. AR sets use quadratic inversion with a $t_{G-1}$ critical value; every disconnected component and unbounded endpoint is shown. Parentheses report standard errors. † denotes an AR set excluding zero, including disconnected sets; it does not imply a bounded estimate or a known effect sign.

**Table 39:** Registration Expansion and Turnout

| | Turnout (percentage points) | | |
|---|---|---|---|
| | (1) | (2) | (3) |
| All registered families | -3.043 | -2.191 | -2.101 |
| | ( 2.037 ) | ( 1.297 ) | ( 1.224 ) |
| AR 95% confidence set | [-13.075, 0.170] | [-6.495, 0.130] | [-5.972, 0.093] |
| Income above R$140 | 9.683 | 23.829 | -9.129 |
| | ( 12.442 ) | ( 61.588 ) | ( 8.783 ) |
| AR 95% confidence set | $(-\infty$, -5.902] $\cup$ [-0.505, $\infty)$ | $(-\infty, \infty)$ | $(-\infty$, 0.584] $\cup$ [6.079, $\infty)$ |
| Income at or below R$140 | -2.253 | -1.779 | -1.981 |
| | ( 1.459 ) | ( 1.033 ) | ( 1.163 ) |
| AR 95% confidence set | [-7.528, 0.123] | [-4.802, 0.104] | [-5.524, 0.086] |
| Income at or below R$70 | -2.386 | -1.916 | -2.299 |
| | ( 1.538 ) | ( 1.113 ) | ( 1.364 ) |
| AR 95% confidence set | [-7.776, 0.130] | [-5.117, 0.111] | [-6.450, 0.098] |
| Income above R$70 and at or below R$140 | -5.534 | -3.703 | -2.523 |
| | ( 4.220 ) | ( 2.455 ) | ( 1.507 ) |
| AR 95% confidence set | [-1280.818, 0.310] | [-36.623, 0.239] | [-9.530, 0.121] |
| Incumbent party and baseline *Bolsa Família* intensity | ✓ | ✓ | ✓ |
| Municipality and mayor characteristics | | ✓ | ✓ |
| Benefit value and *FPM* transfers | | | ✓ |
| Number of Municipalities | 2,078 | 2,078 | 2,078 |
| Observations | 4,156 | 4,156 | 4,156 |

*Notes:* All columns retain the locked 2008–2012 sample, quota-shock instrument, municipality and year fixed effects, 2006-population weights and state-clustered CRV1 standard errors. Controls are interacted with post. Registration expansion counts families first registered in 2009–2012 and still observed in the 2012 *Cadastro Único* snapshot. Income (monthly per capita) and municipality are measured in 2012. These are surviving entry cohorts, not historical net stocks or verified *Bolsa Família* recipients. The income definitions overlap: the at-or-below-R$140 group combines the two lower-income bands. Each row uses a separate endogenous registration measure, scaled by its own 2006-population-weighted SD of the change in registrations per 10,000 residents. Coefficients are not directly comparable across row-specific SDs. One common instrument does not identify separate causal effects by family income or the effect of registration apart from subsequent transfers. First-stage diagnostics appear in Table 36. AR sets use quadratic inversion with a $t_{G-1}$ critical value; every disconnected component and unbounded endpoint is shown. Parentheses report standard errors. † denotes an AR set excluding zero, including disconnected sets; it does not imply a bounded estimate or a known effect sign.

**Table 40:** Quota Effects on Registration, Access, and Votes by CRAS Presence

| Outcome | No CRAS<br>(1) | CRAS<br>(2) | Difference<br>(3) | Mean<br>(4) |
|---|---|---|---|---|
| All registrations | 45.367 (10.695)<br>[23.340, 67.394] | 37.932 (11.644)<br>[13.951, 61.913] | -7.435 (13.950)<br>[-36.166, 21.295]<br>Holm p: 1.000 | 1624.362 |
| Registrations: income at most R$140 | 42.995 (6.473)<br>[29.665, 56.326] | 34.808 (9.916)<br>[14.386, 55.231] | -8.187 (10.842)<br>[-30.517, 14.143]<br>Holm p: 1.000 | 1225.048 |
| Registrations: income above R$140 | 2.372 (5.854)<br>[-9.685, 14.429] | 3.124 (3.896)<br>[-4.901, 11.148] | 0.752 (5.607)<br>[-10.797, 12.300]<br>Holm p: 1.000 | 399.314 |
| Beneficiary families | 58.878 (7.206)<br>[44.036, 73.719] | 54.639 (10.916)<br>[32.157, 77.121] | -4.239 (10.212)<br>[-25.270, 16.793]<br>Holm p: 1.000 | 981.615 |
| Monthly payments per resident (R$) | 0.920 (0.119)<br>[0.675, 1.165] | 0.887 (0.172)<br>[0.532, 1.242] | -0.033 (0.165)<br>[-0.374, 0.307]<br>Holm p: 1.000 | 10.807 |
| Incumbent vote share (pp) | -0.398 (0.634)<br>[-1.705, 0.909] | 1.841 (0.540)<br>[0.729, 2.953] | 2.239 (0.736)<br>[0.724, 3.755]<br>Holm p: 0.229 | 51.456 |
| Turnout (pp) | -0.138 (0.073)<br>[-0.288, 0.012] | -0.385 (0.213)<br>[-0.823, 0.054] | -0.247 (0.206)<br>[-0.671, 0.178]<br>Holm p: 1.000 | 87.264 |
| Number of Mayors / Municipalities | 684 | 1394 | 2078 | |
| Observations | 1368 | 2788 | 4156 | |

*Notes:* All estimates use one full-sample population-weighted SD of the authorized quota shock (171.496 families per 10,000 residents). Weights are 2006 population; municipality/year effects and the complete baseline specification are included. Parentheses: state-clustered SEs; brackets: 95% CRV1 t intervals. Holm p uses 9,999 null-imposed state wild-cluster draws. Registrations are retrospective cohorts surviving in the 2012 snapshot, classified by 2012 income, not verified PBF receipt. All baseline controls, party trends and post effects vary by group. Difference is second minus first; seven outcome contrasts form one Holm family. Family quantities are per 10,000 residents. Beneficiary families and nominal monthly payments are calendar-year averages; payments combine quantity and generosity. Mean is unweighted over the two-period sample. A weak contrast does not establish equivalence. CRAS presence predates the reform; comparisons do not identify the effect of installing an office.

**Table 41:** Quota Effects by Verified Active Outreach within CRAS Municipalities

| Outcome | No outreach (1) | Outreach (2) | Difference (3) | Mean (4) |
|---|---|---|---|---|
| All registrations | 63.992 (23.464)<br>[15.668, 112.316] | 33.290 (8.396)<br>[15.999, 50.582] | -30.702 (20.651)<br>[-73.234, 11.829]<br>Holm p: 1.000 | 1731.691 |
| Registrations: income at most R$140 | 60.996 (20.125)<br>[19.549, 102.443] | 29.655 (6.372)<br>[16.531, 42.779] | -31.341 (17.075)<br>[-66.508, 3.827]<br>Holm p: 1.000 | 1331.899 |
| Registrations: income above R$140 | 2.996 (4.895)<br>[-7.085, 13.077] | 3.635 (4.474)<br>[-5.579, 12.849] | 0.638 (5.746)<br>[-11.196, 12.473]<br>Holm p: 1.000 | 399.792 |
| Beneficiary families | 79.014 (19.614)<br>[38.619, 119.408] | 49.968 (7.608)<br>[34.298, 65.637] | -29.046 (15.287)<br>[-60.530, 2.438]<br>Holm p: 1.000 | 1080.183 |
| Monthly payments per resident (R$) | 1.249 (0.301)<br>[0.628, 1.869] | 0.813 (0.126)<br>[0.552, 1.073] | -0.436 (0.232)<br>[-0.915, 0.043]<br>Holm p: 1.000 | 12.067 |
| Incumbent vote share (pp) | 0.757 (0.742)<br>[-0.771, 2.285] | 2.004 (0.646)<br>[0.673, 3.334] | 1.247 (0.897)<br>[-0.601, 3.094]<br>Holm p: 1.000 | 51.362 |
| Turnout (pp) | -0.234 (0.167)<br>[-0.579, 0.110] | -0.394 (0.243)<br>[-0.894, 0.106] | -0.160 (0.155)<br>[-0.478, 0.159]<br>Holm p: 1.000 | 86.175 |
| Number of Mayors / Municipalities | 377 | 1017 | 1394 | |
| Observations | 754 | 2034 | 2788 | |

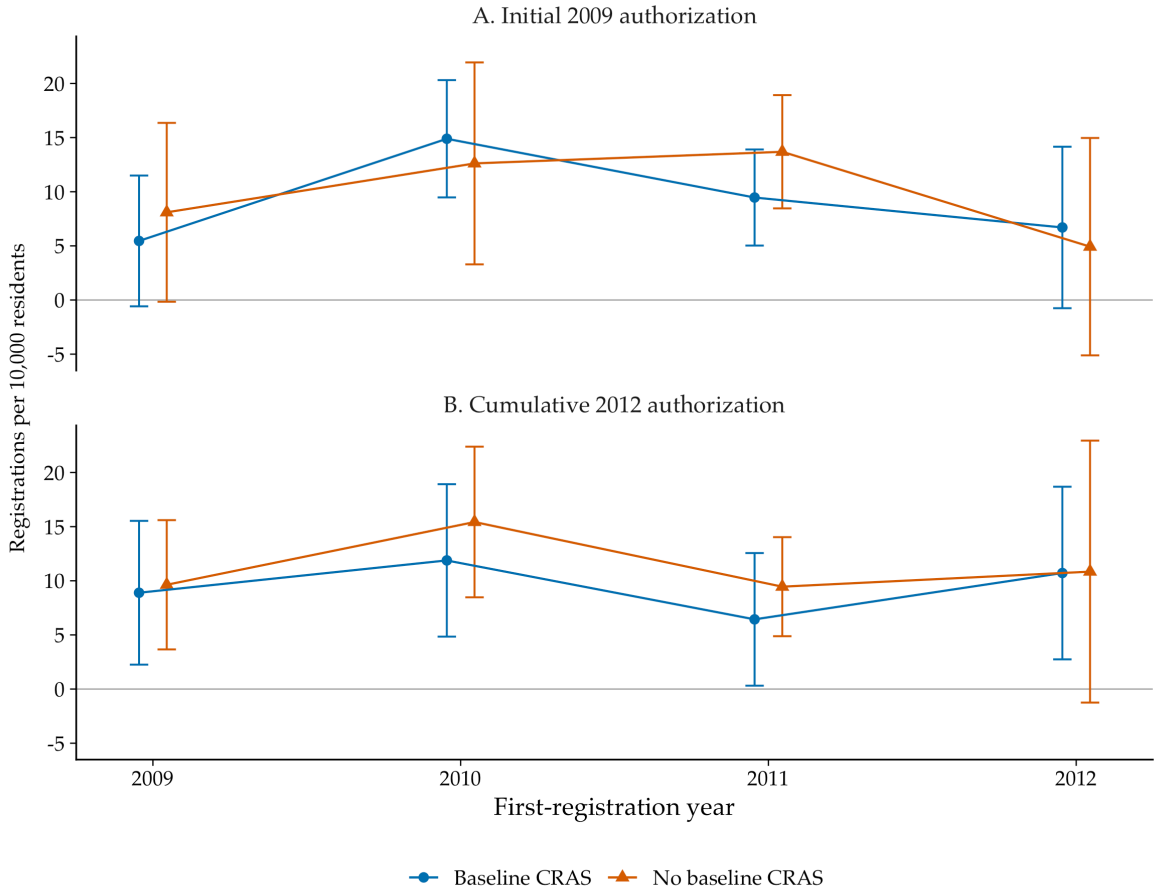
*Notes:* All estimates use one full-sample population-weighted SD of the authorized quota shock (171.496 families per 10,000 residents). Weights are 2006 population; municipality/year effects and the complete baseline specification are included. Parentheses: state-clustered SEs; brackets: 95% CRV1 t intervals. Holm p uses 9,999 null-imposed state wild-cluster draws. Registrations are retrospective cohorts surviving in the 2012 snapshot, classified by 2012 income, not verified PBF receipt. All baseline controls, party trends and post effects vary by group. Difference is second minus first; seven outcome contrasts form one Holm family. Family quantities are per 10,000 residents. Beneficiary families and nominal monthly payments are calendar-year averages; payments combine quantity and generosity. Mean is unweighted over the two-period sample. A weak contrast does not establish equivalence. Outreach is verified from literal 2008 Censo SUAS yes/no reports; municipalities without CRAS are excluded.

**Table 42:** Quota-Related Registration Cohorts by Baseline CRAS

| Cohort | No CRAS | CRAS | Difference | 95% interval | Holm $p$ |
|---|---|---|---|---|---|
| *Panel A: Initial 2009 authorization* | | | | | |
| 2009 | 8.099 | 5.456 | -2.642 | [-12.930, 7.645] | 1.000 |
| 2010 | 12.618 | 14.892 | 2.274 | [-5.820, 10.369] | 1.000 |
| 2011 | 13.693 | 9.466 | -4.227 | [-10.150, 1.695] | 1.000 |
| 2012 | 4.926 | 6.700 | 1.774 | [-9.597, 13.145] | 1.000 |
| *Panel B: Cumulative 2012 authorization* | | | | | |
| 2009 | 9.632 | 8.895 | -0.737 | [-8.087, 6.614] | 1.000 |
| 2010 | 15.431 | 11.883 | -3.548 | [-13.544, 6.448] | 1.000 |
| 2011 | 9.456 | 6.438 | -3.018 | [-7.193, 1.157] | 1.000 |
| 2012 | 10.849 | 10.716 | -0.133 | [-13.803, 13.538] | 1.000 |

Notes: The unchanged headline sample contains 2,078 municipalities and 26 states. Each outcome counts families first registered in the indicated year and retained in the 2012 snapshot, per 10,000 residents using fixed 2006 population. Coefficients are responses to one full-sample population-weighted SD of the specified authorization shock. Controls and year effects are fully interacted with baseline CRAS; municipality effects, 2006-population weights and state clustering follow the main design. Difference is CRAS minus no CRAS. Intervals use the clustered covariance and a $t_{25}$ reference. Holm p-values adjust 9,999-draw state wild-bootstrap tests across the eight displayed contrasts. Panel B partitions the main first-stage response retrospectively: it does not show that 2012 authorization caused registrations in earlier years. Instruments differ in scale and are estimated separately. No monthly pre-2009 registration trend is observed in these aggregates; timing alone does not identify electoral effort.

**Figure 9:** Annual Registration Responses by Baseline CRAS




*Notes:* Each point is the quota-related response of the indicated registration cohort, per 10,000 residents, for municipalities with or without a CRAS before the reform. The two panels use separate authorization measures, each scaled by its own full-sample population-weighted standard deviation. Bars are pointwise state-clustered 95% confidence intervals. Table 42 reports the group contrasts and multiplicity-adjusted bootstrap tests. Families are observed retrospectively in the 2012 snapshot. Panel B decomposes the cumulative main response by registration year; it does not imply that a future quota change caused earlier registrations. The available aggregates cannot identify pre-2009 monthly trends or families that exited the registry.